# Data Structures

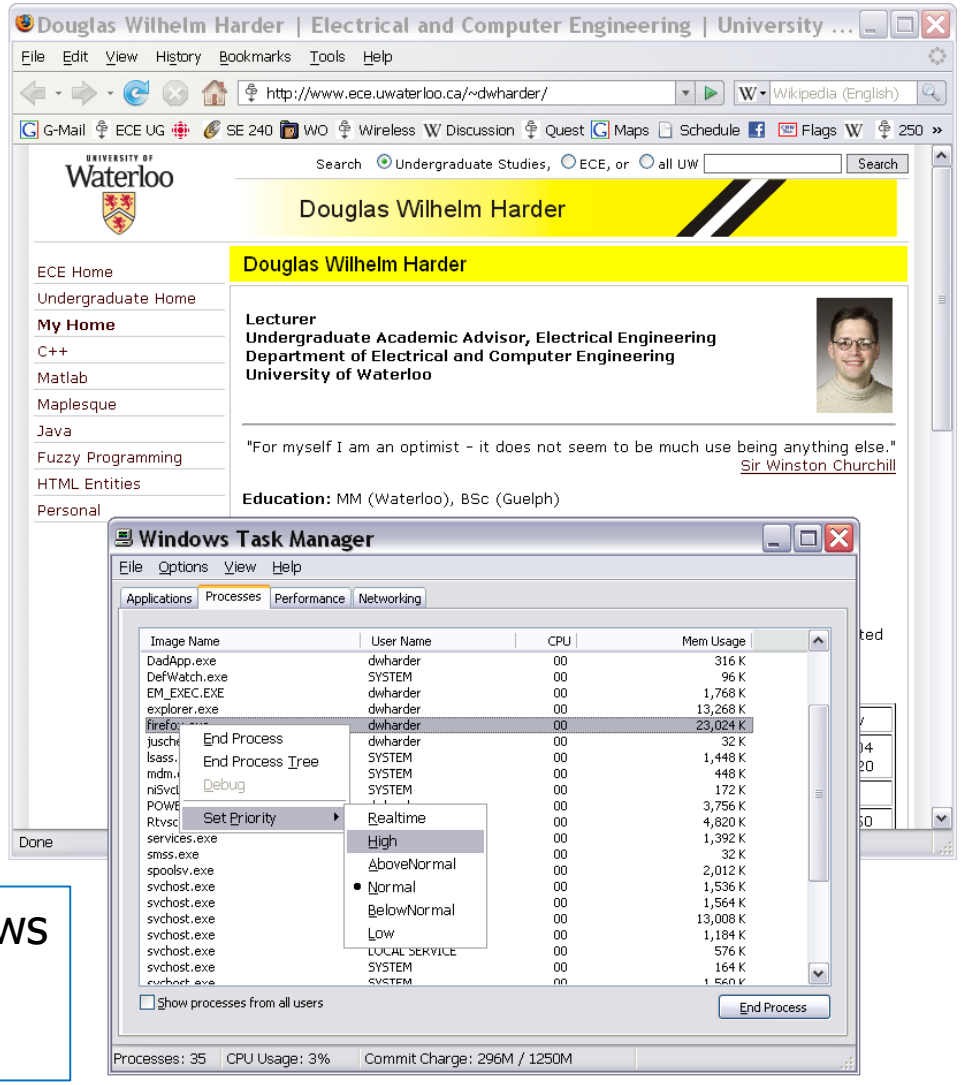
## 14. Heap (Priority Queues)

# Motivation

- With queues, the order may be summarized by first in, first out
- Some tasks may be more important or timely than others
  - Higher priority
- Priority queues
  - Enqueue objects using a partial ordering based on priority
  - Dequeue that object which has the highest priority

# Applications Of Priority Queue

- Hold jobs for a printer in order of length
- Store packets on network routers in order of urgency
- Ordering CPU jobs
- Emergency room admission processing




The priority of processes in Windows may be set in the Windows Task Manager

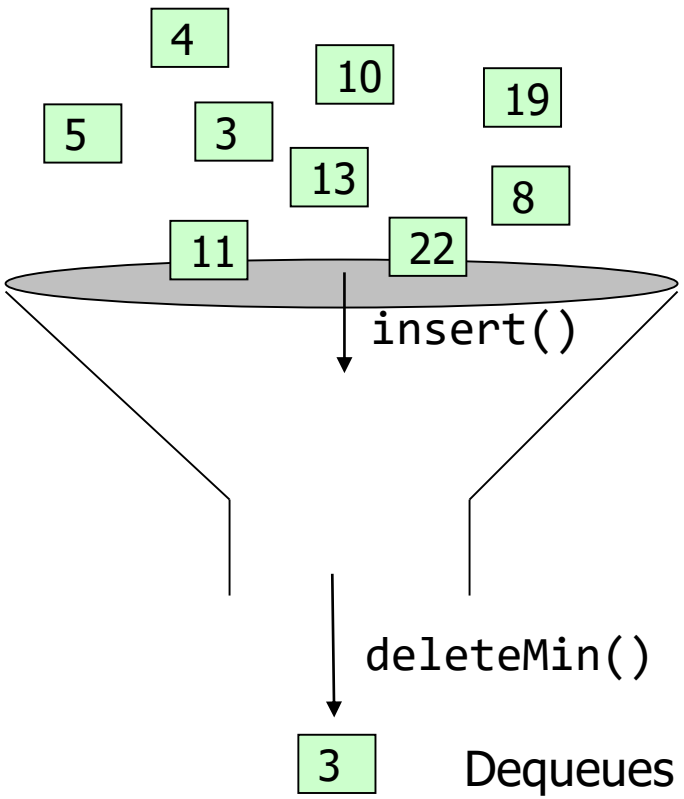
# Priority Queue – ADT

- `insert` (i.e., enqueue)
  - Dynamic insert
  - Specification of a priority level (0-high, 1,2.. Low)

- `deleteMin` (i.e., dequeue)
  - Returns the current "highest priority" element in the queue
    - Element with the minimum priority level
  - Deletes that element from the queue

- Performance goal is to make the run time of each operation as close to `O(1)` as possible

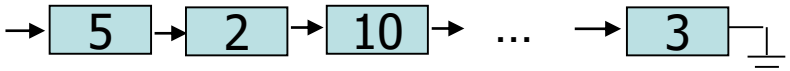
# Priority Queue – ADT



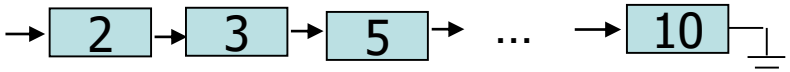

Dequeues the next element with the highest priority

# Simple Implementations

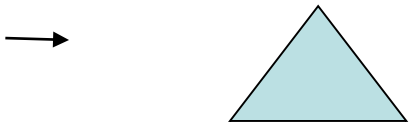
- Unordered linked list
  - `Insert` – `O(1)` step
  - `deleteMin` – `O(n)` steps




- Ordered linked list
  - `insert` – `O(n)` steps
  - `deleteMin` – `O(1)` step




- Balanced binary tree, e.g., AVL Tree
  - `insert` – $O(\log_2 n)$ steps
  - `deleteMin` in how many steps?
    - Find min – $O(\log_2 n)$ steps
    - Delete – $O(\log_2 n)$ steps



Can we build a data structure better suited to store and retrieve priorities?

# Binary Heap

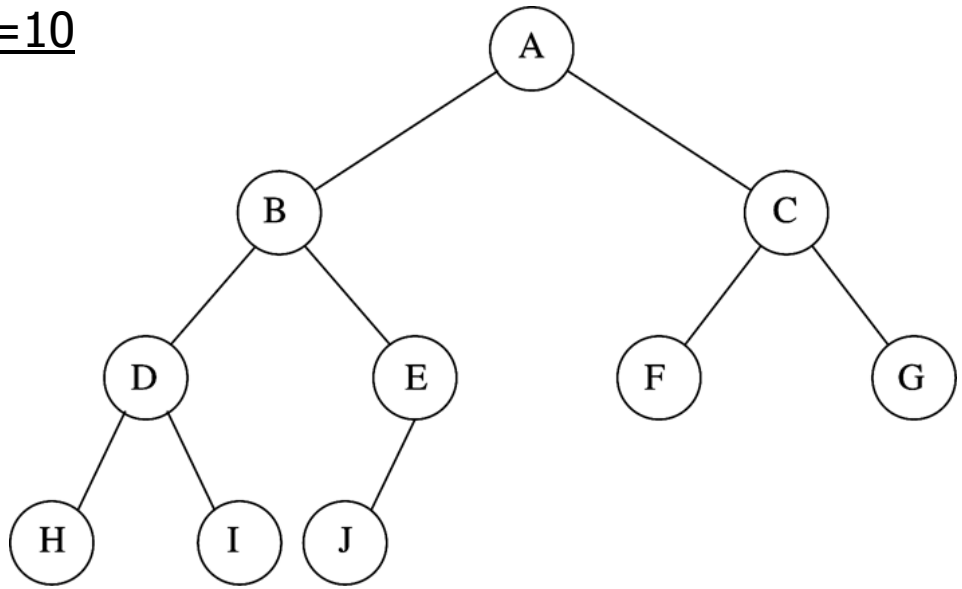
# Binary Heap

- A binary heap is a binary tree with two properties
  - Structure property
  - Heap-order property

# Binary Heap – Structure Property

- A binary heap is (almost) complete binary tree
  - Each level (except possibly the bottom most level) is completely filled
  - The bottom most level may be partially filled (from left to right)

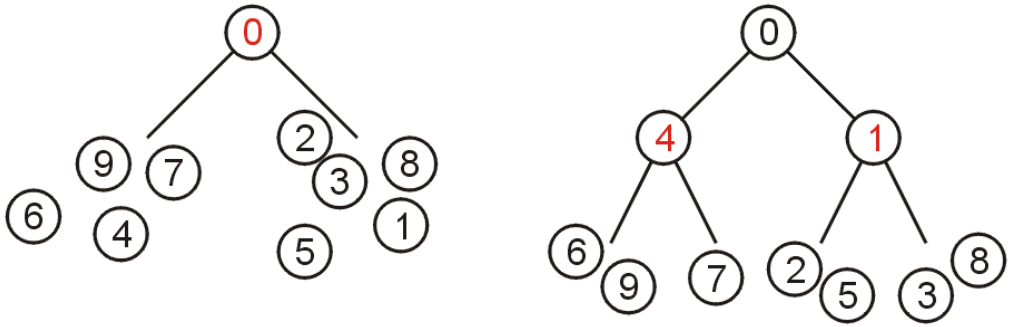
N=10




Every level (except last) saturated

# Binary Heap – Heap-Order Property

- Min-Heap property
  - Key associated with the root is less than or equal to the keys associated with either of the sub-trees (if any)
  - Both of the sub-trees (if any) are also binary min-heaps




- Properties of min-heap
  - A single node is a min-heap
  - Minimum key always at root
  - For every node X, `key(parent(X)) ≤ key(X)`
  - No relationship between nodes with similar key

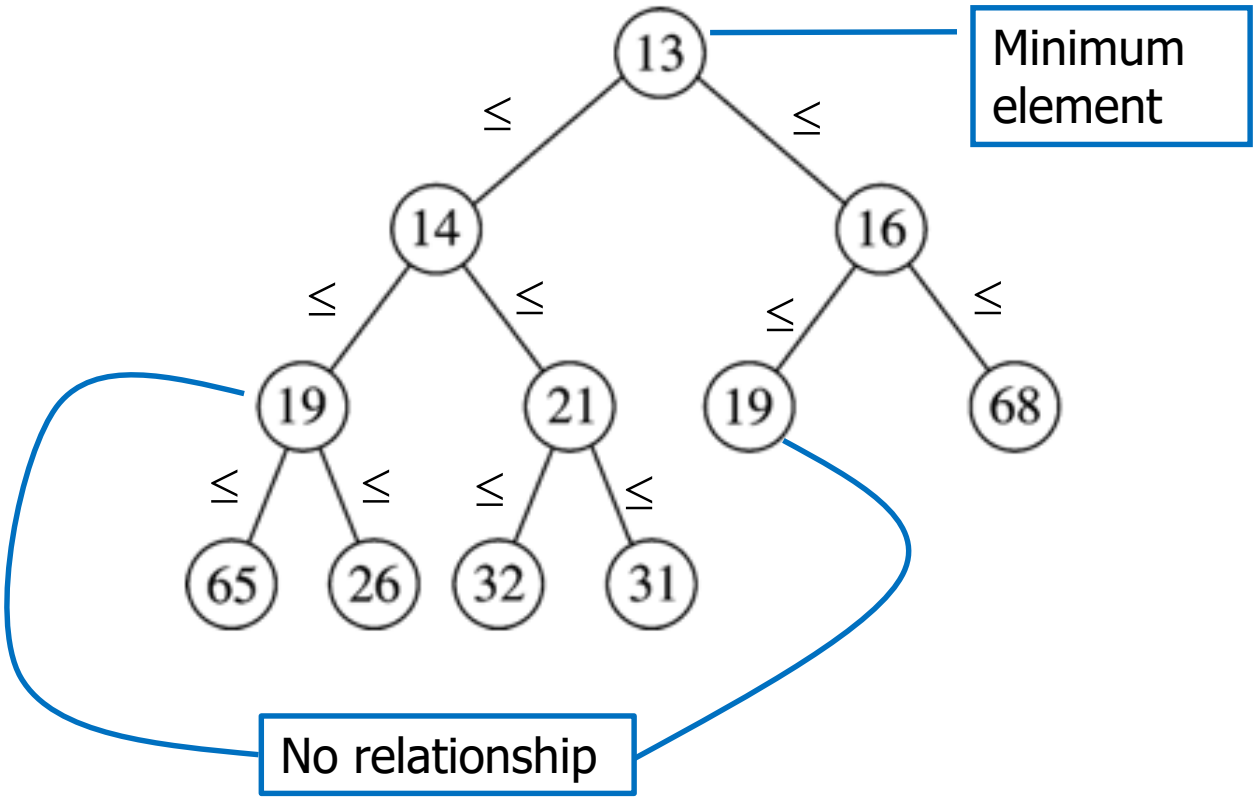
# Binary Heap – Heap-Order Property

- Max-Heap property
  - Maximum key at the root
  - For every node `X`, `key(parent(X)) ≥ key(X)`

- `Insert` and `deleteMin` must maintain heap-order property

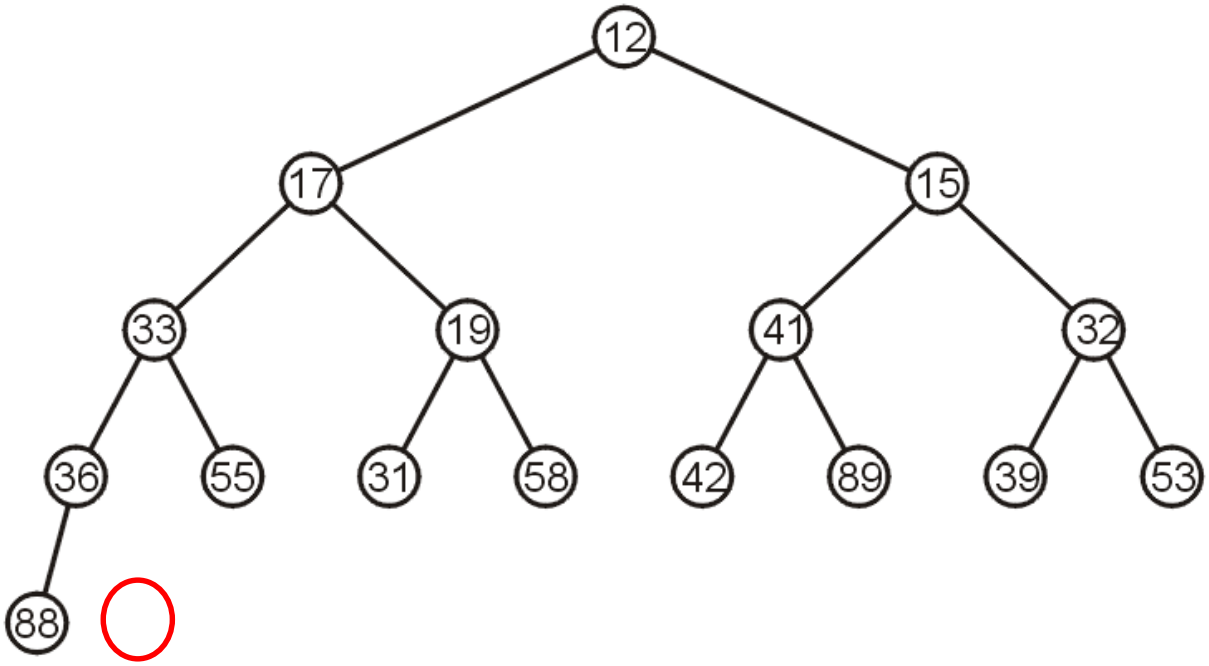
# Heap-Order Property – Example

- Min-Heap



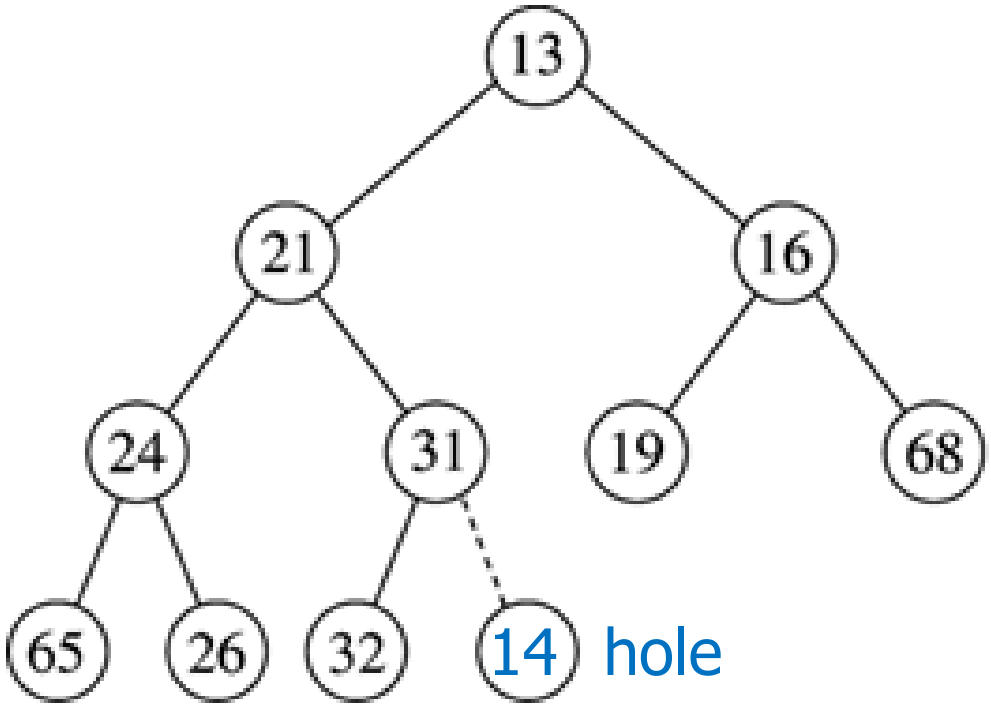
# Heap Operations – `insert`

- Insert new element into the heap at the next available slot ("hole")
  - Maintaining (almost) complete binary tree
- Percolate the element up the heap while heap-order property not satisfied



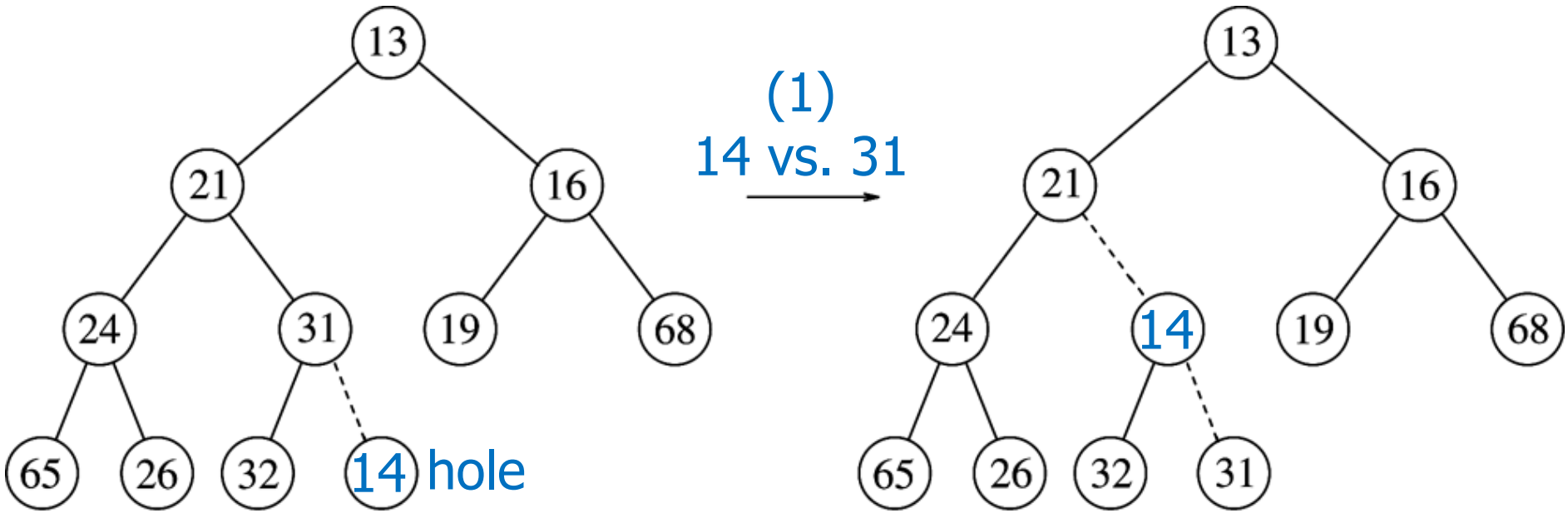
# Heap Insert – Example

- Insert 14



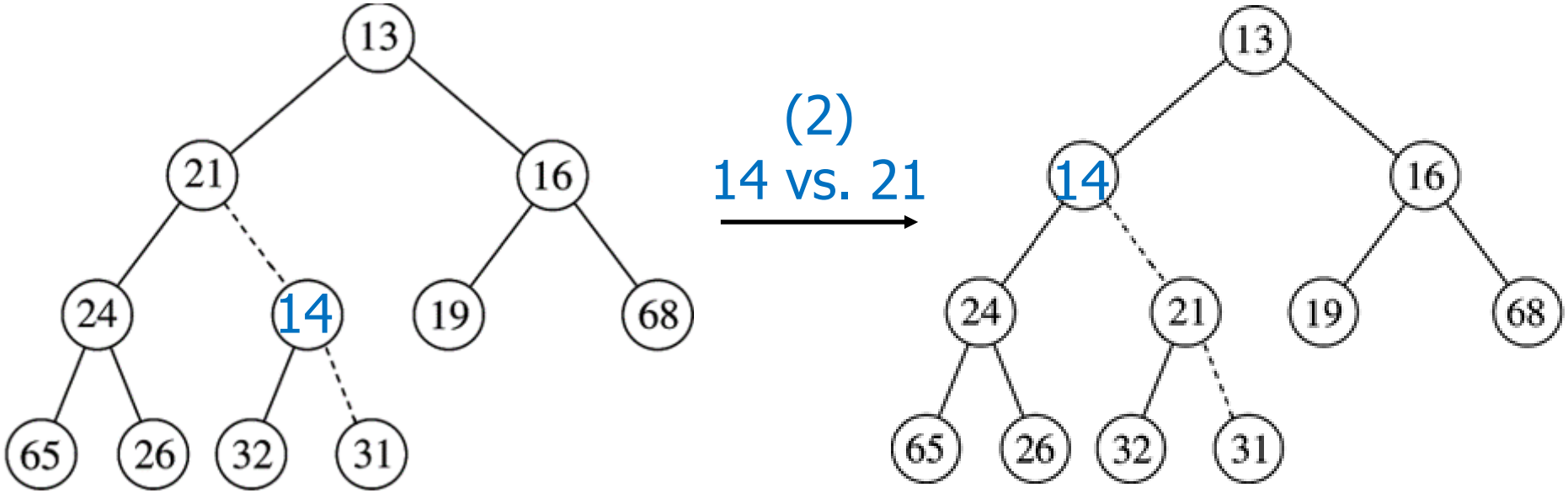
# Heap Insert – Example

- Insert 14



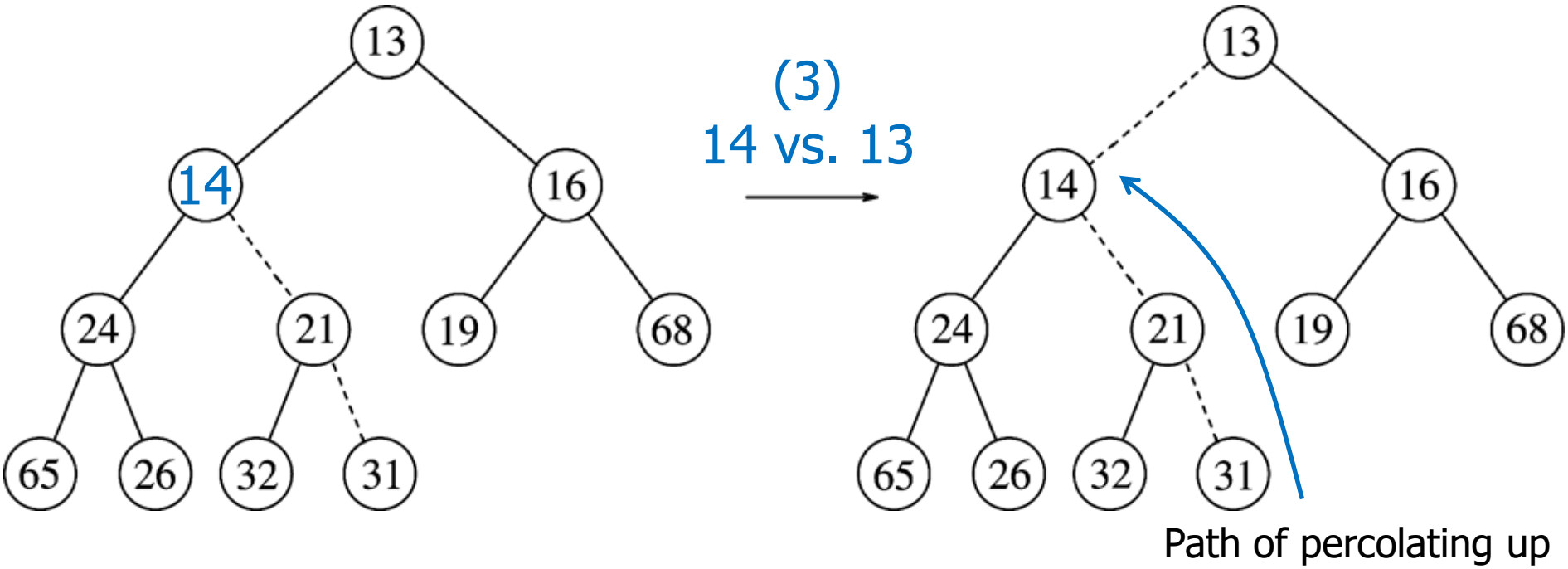
# Heap Insert – Example

- Insert 14



# Heap Insert – Example

- Insert 14

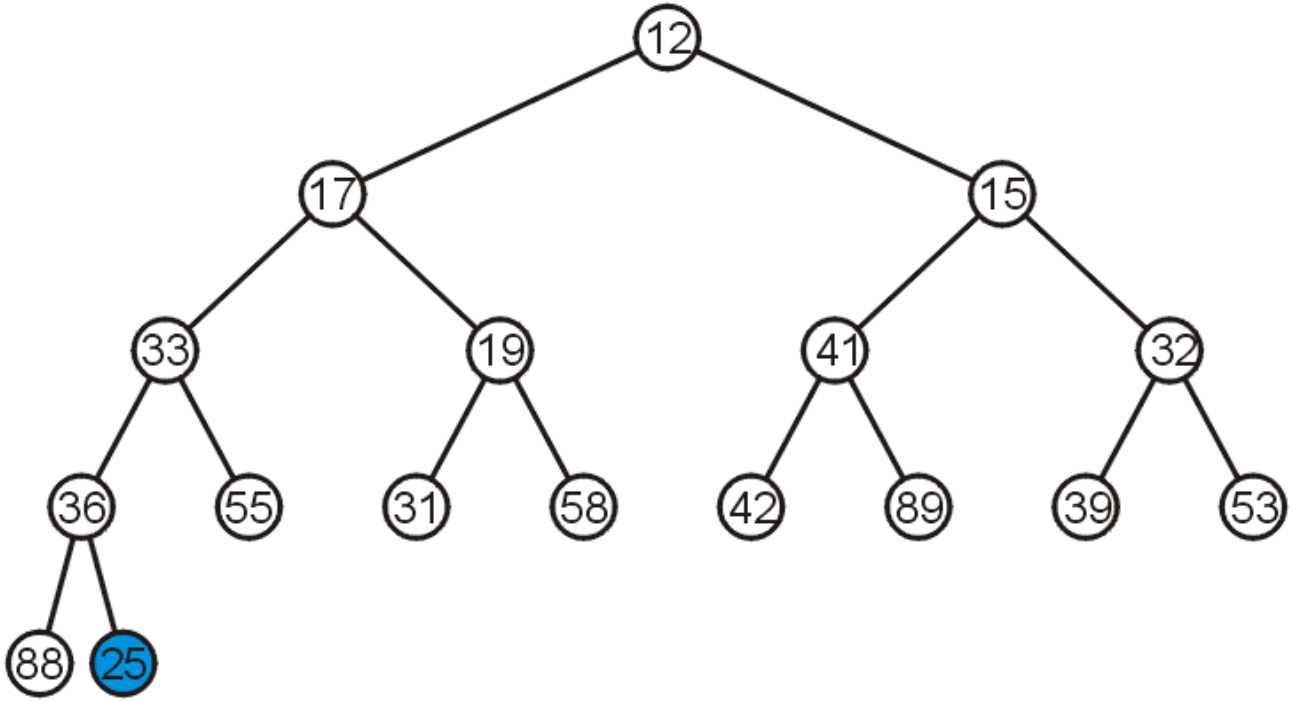



✓ Heap order property

✓ Structure property

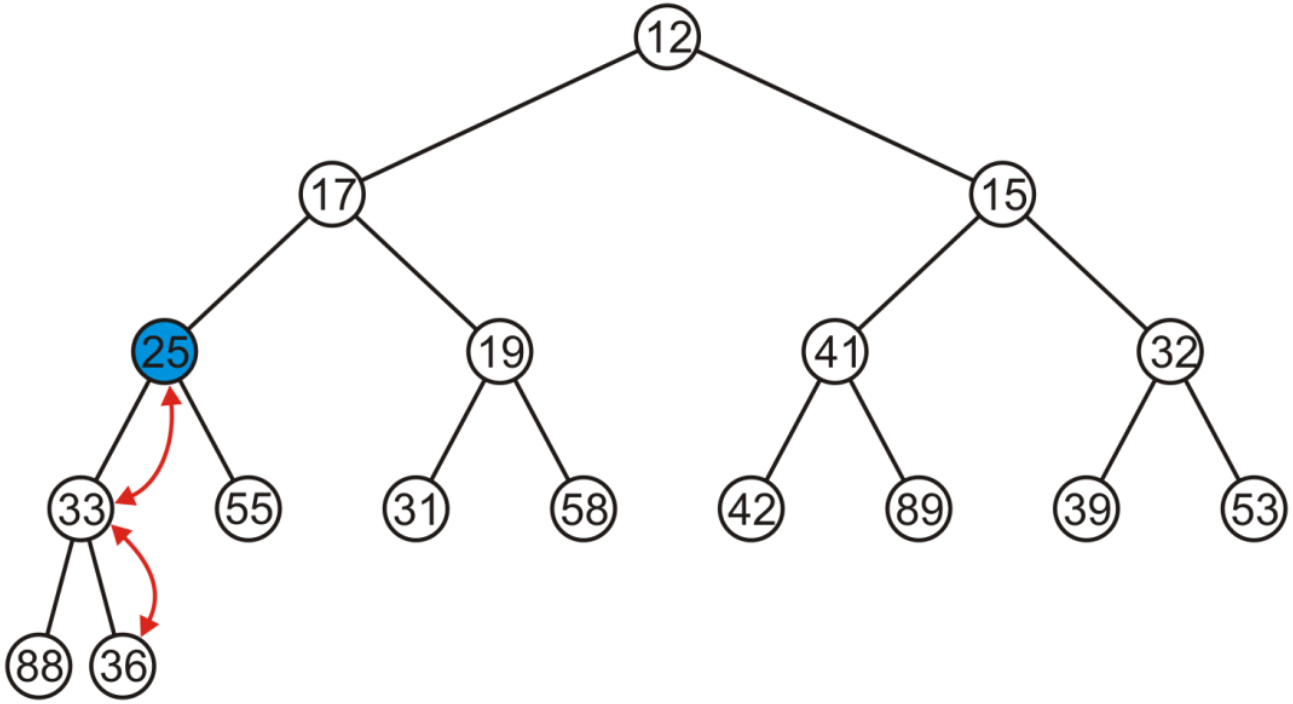
# Heap Insert – Example

- Insert 25



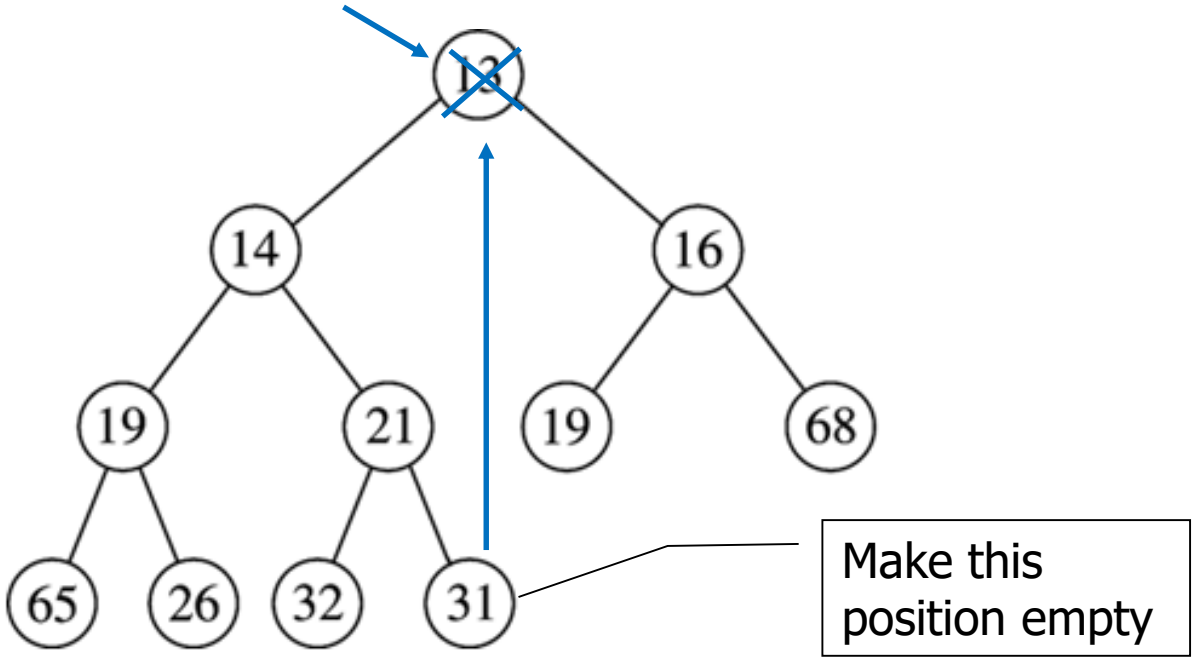
# Heap Insert – Example

- Percolate 25 up into its appropriate location
  - The resulting heap is still a complete tree

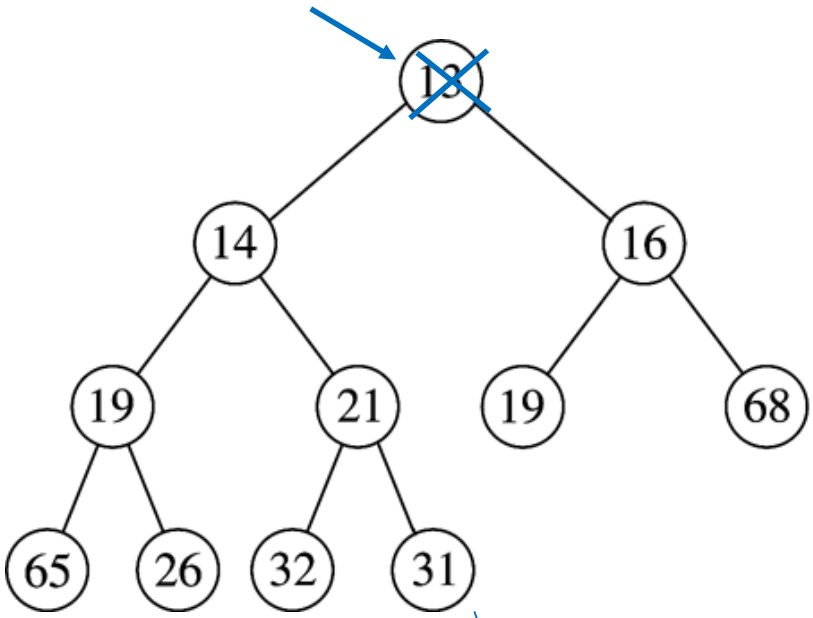


# Heap Operation – `deleteMin`

- Minimum element is always at the root
  - Return the element at the root and delete it
- Heap decreases by one in size
- Move last element of the tree into hole at root
- Percolate down while heap-order property not satisfied

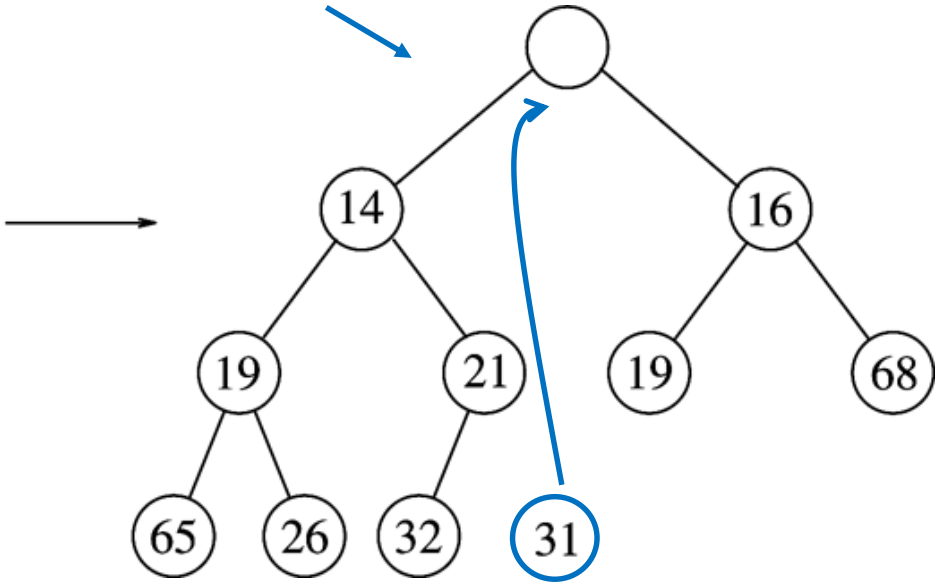


# deleteMin – Example




Make this position empty

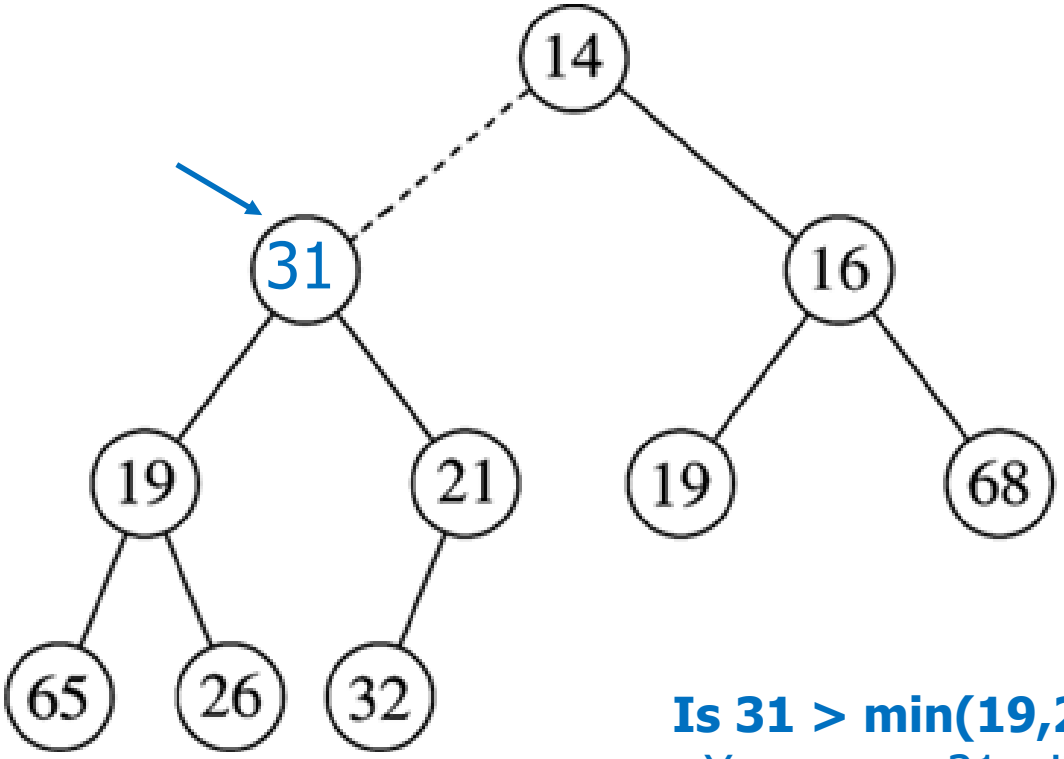
Copy 31 temporarily here and move it down




**Is 31 > min(14,16)?**

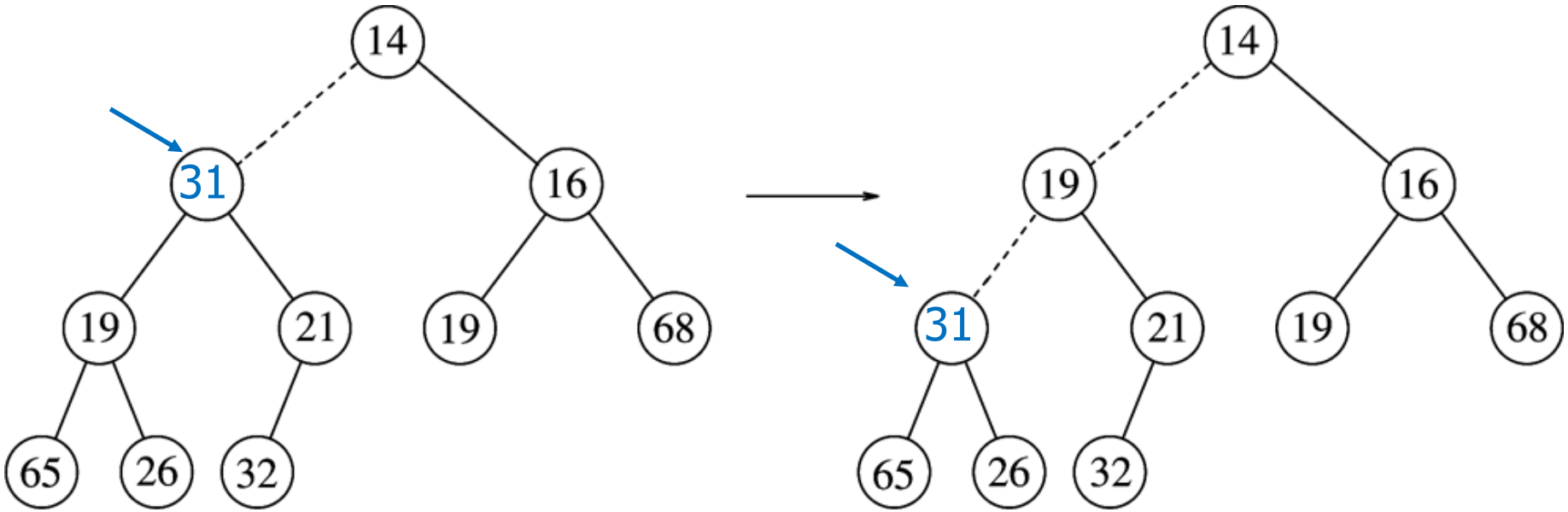
- Yes - swap 31 with min(14,16)

# deleteMin – Example




**Is 31 > min(19,21)?**
- Yes - swap 31 with min(19,21)

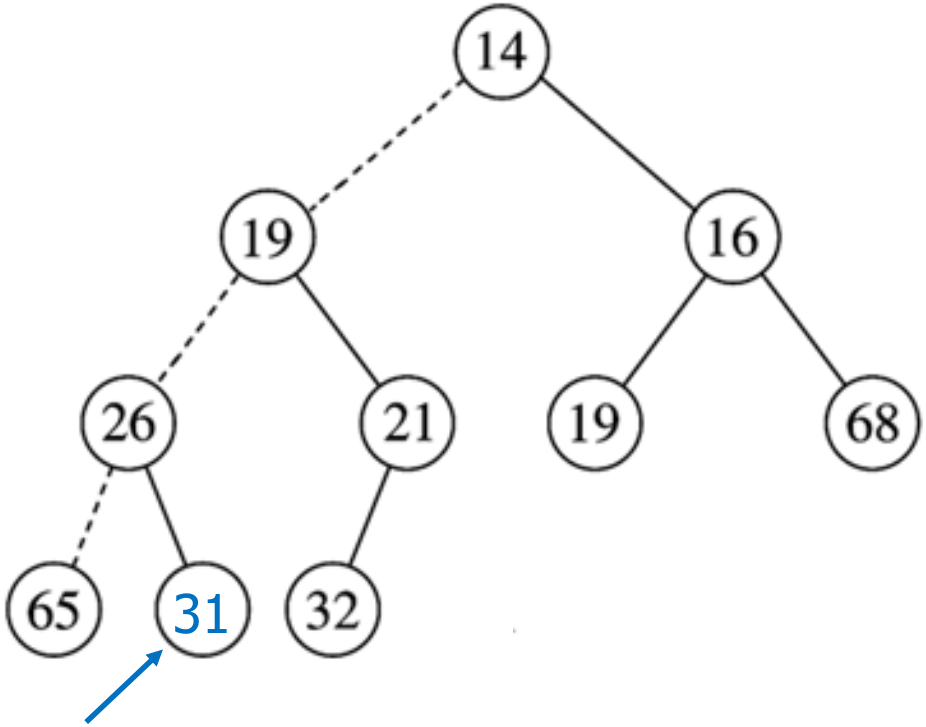
# deleteMin – Example




**Is 31 > min(19,21)?**
- Yes - swap 31 with min(19,21)

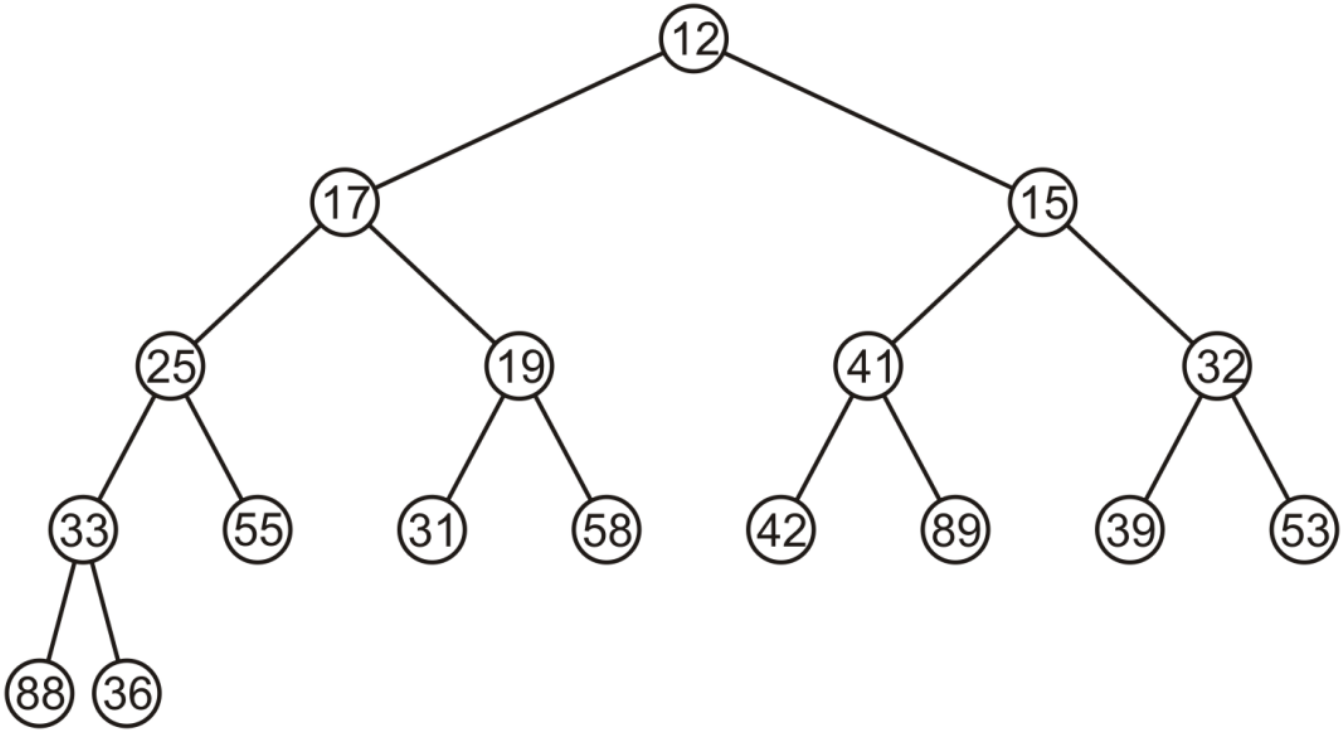
**Is 31 > min(65,26)?**
- Yes - swap 31 with min(65,26)

# deleteMin – Example



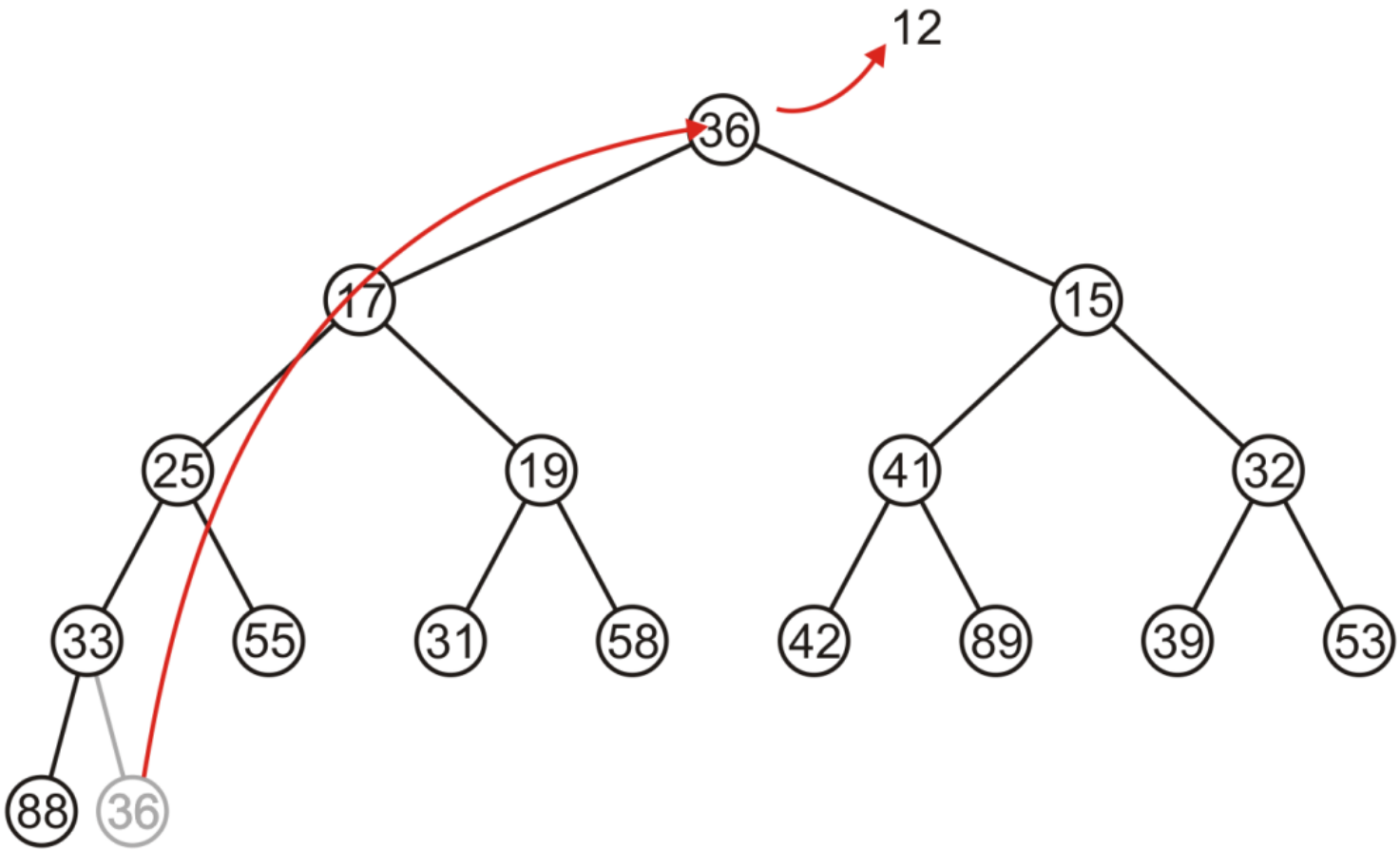
# `deleteMin` – Example

- `deleteMin` will dequeue element 12 from the top



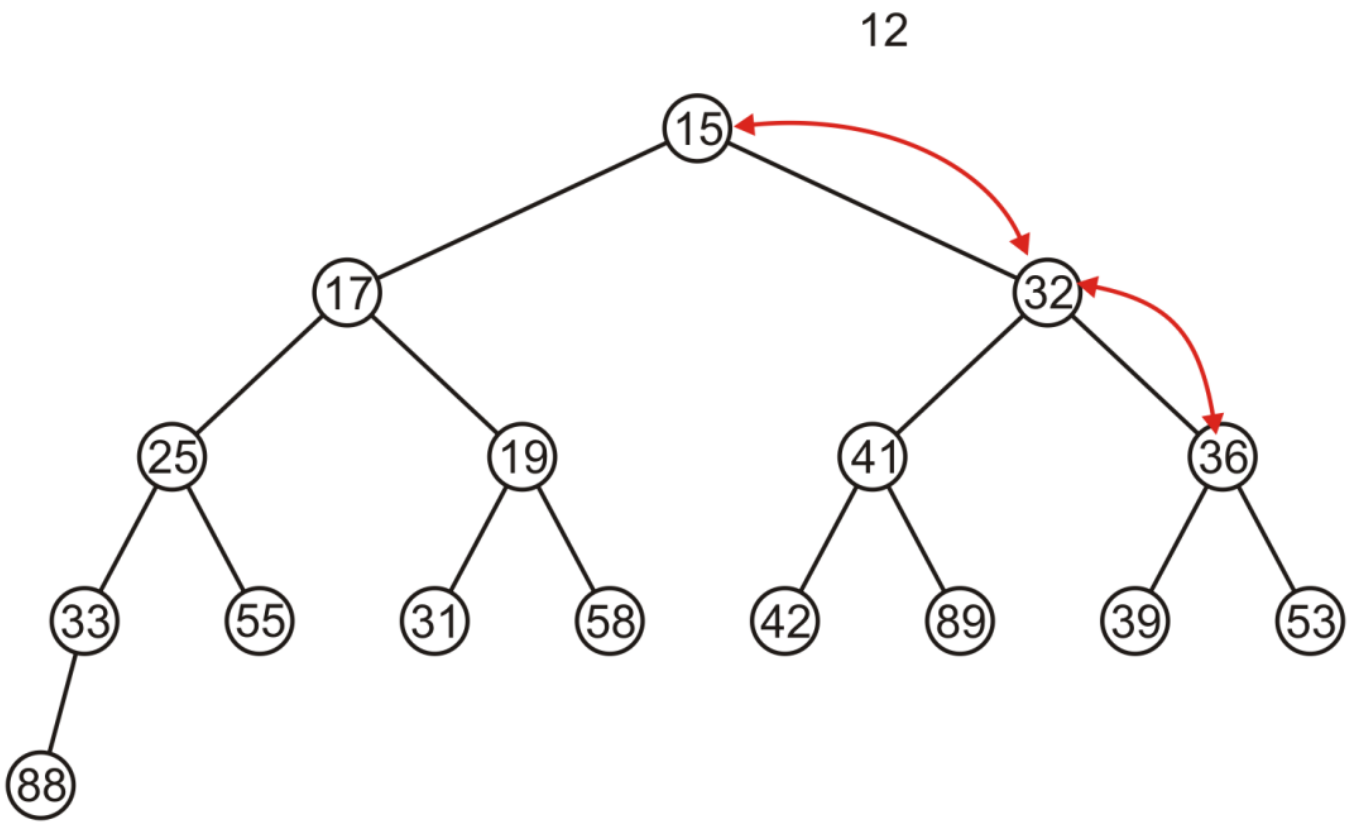
# deleteMin – Example

- Copy the last entry in the heap to the root



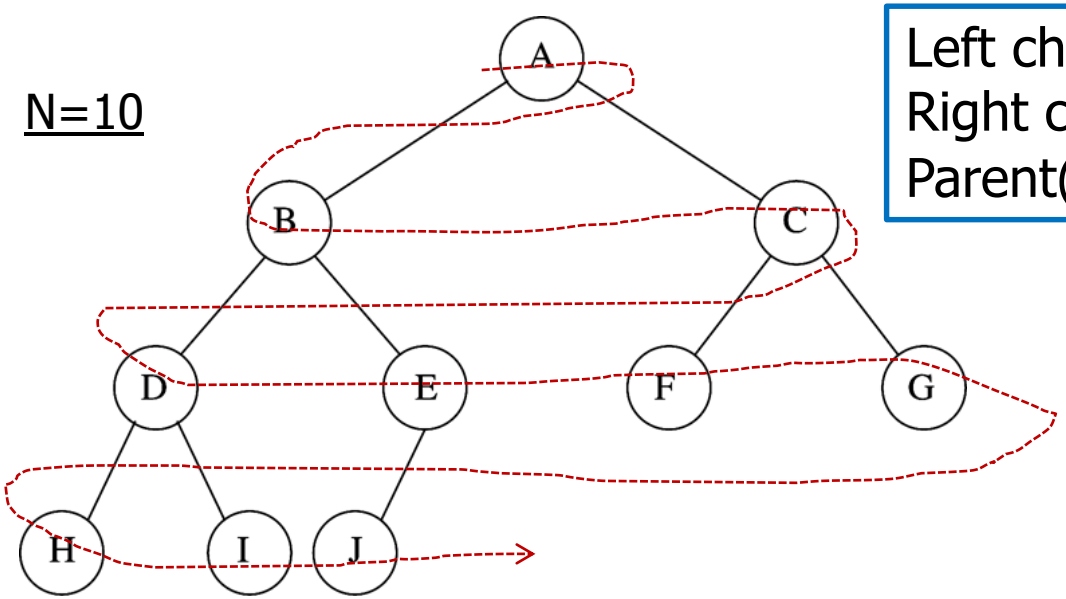
# deleteMin – Example

- Percolate 36 down swapping it with the smallest of its children
  - Halt when both children are larger



# Array-Based Implementation Of Binary Tree

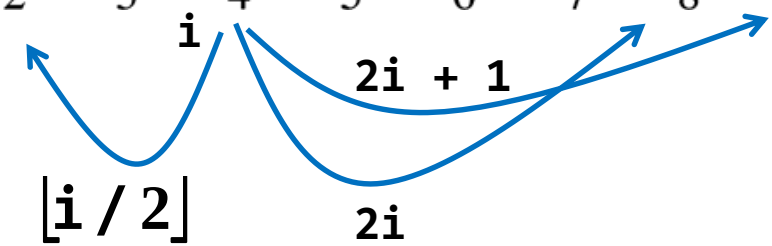
N=10




Left child(i) = at position `2i`
Right child(i) = at position `2i + 1`
Parent(i) = at position $\lfloor i / 2 \rfloor$

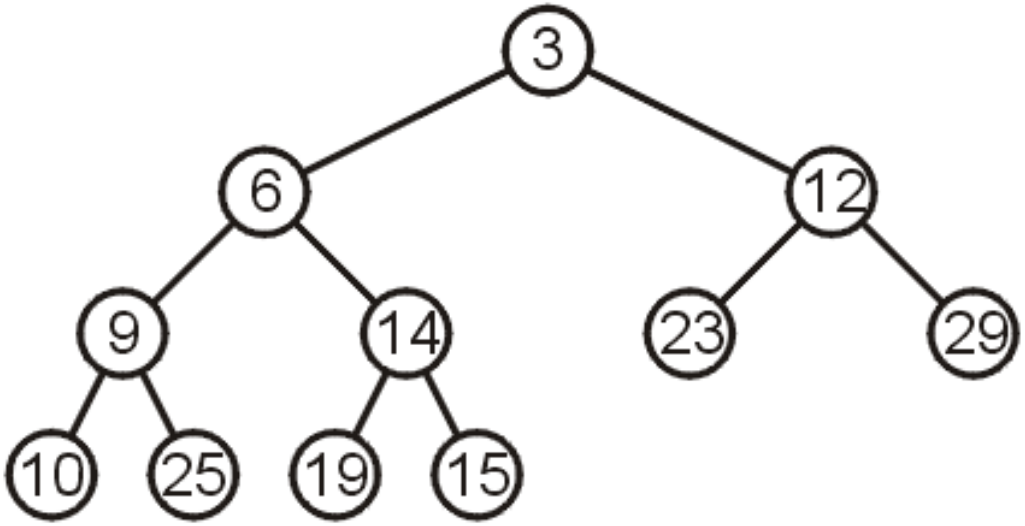
Array representation:

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A | B | C | D | E | F | G | H | I | J | | | |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |



# Array-Based Implementation Of Binary Heap

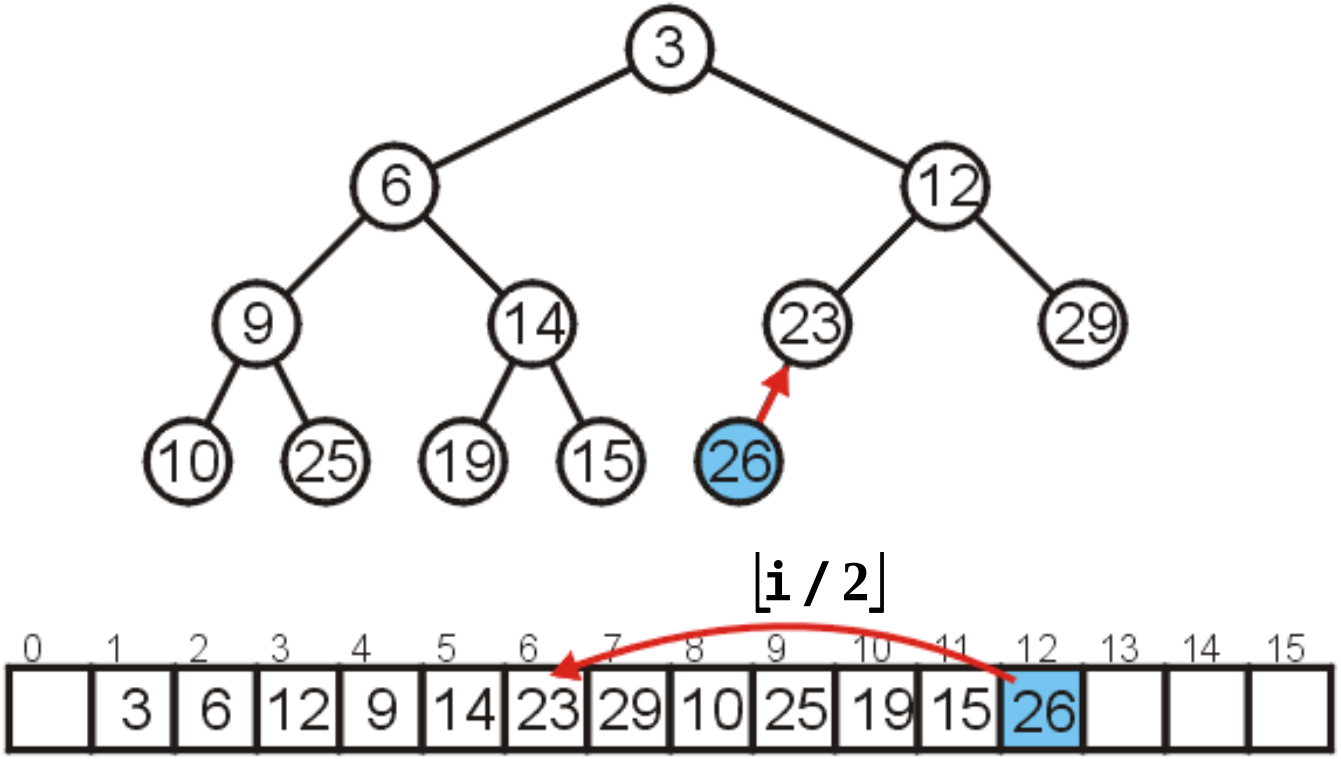
- Consider the following heap, both as a tree and in its array representation




| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 3 | 6 | 12 | 9 | 14 | 23 | 29 | 10 | 25 | 19 | 15 | | | | |

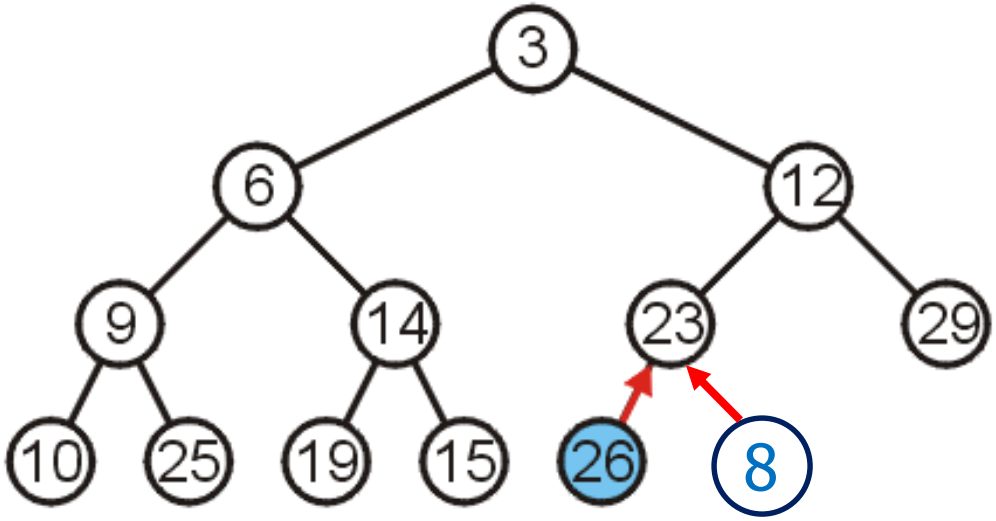
# Array-Based Implementation – `insert`

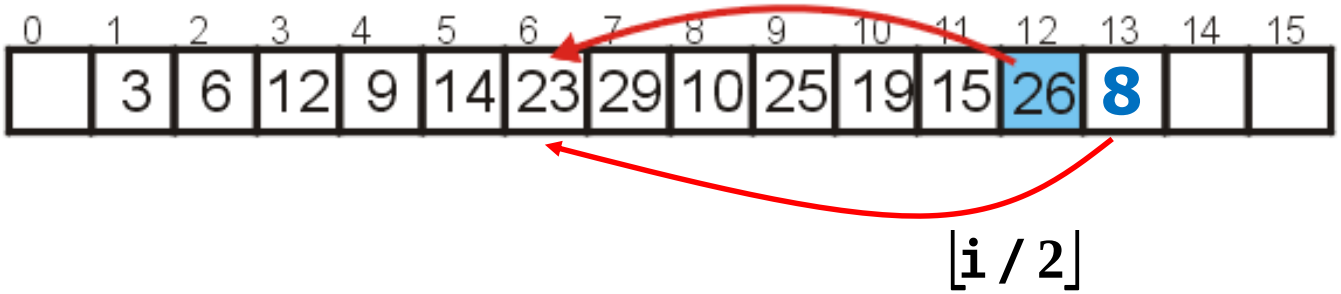
- Inserting 26 requires no changes



# Array-Based Implementation – `insert`

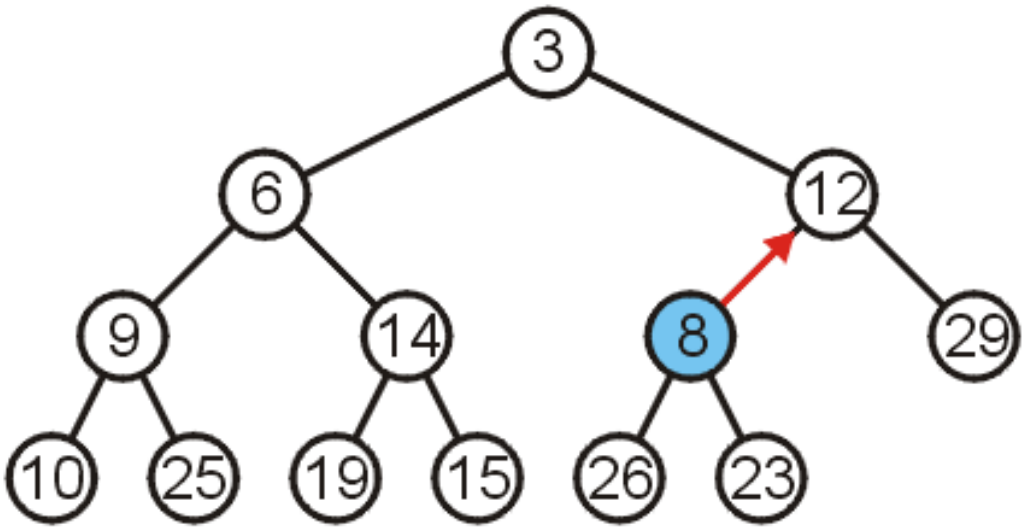
- Inserting 8 requires a few percolations
  - Swap 8 and 23



# Array-Based Implementation – `insert`

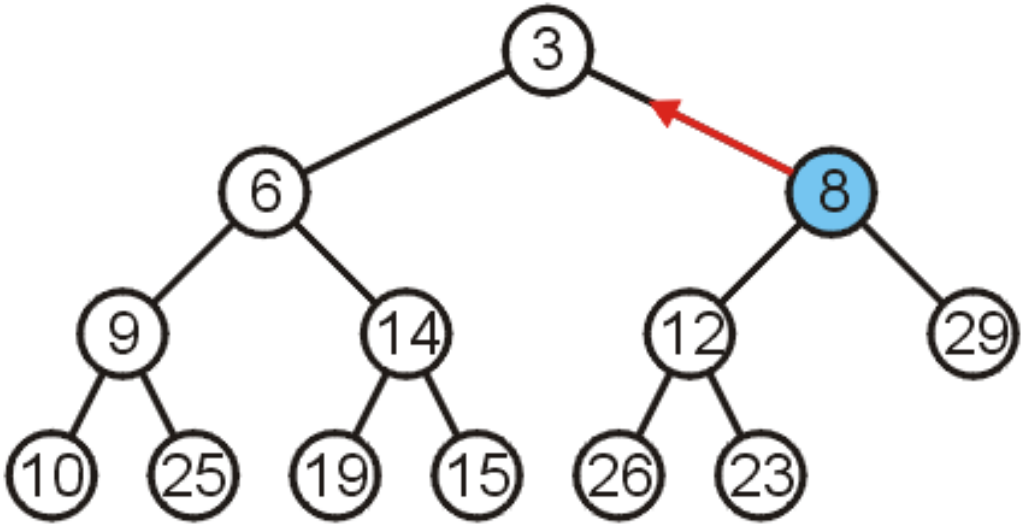
- Swap 8 and 12




| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 3 | 6 | 12 | 9 | 14 | 8 | 29 | 10 | 25 | 19 | 15 | 26 | 23 | | |

# Array-Based Implementation – `insert`

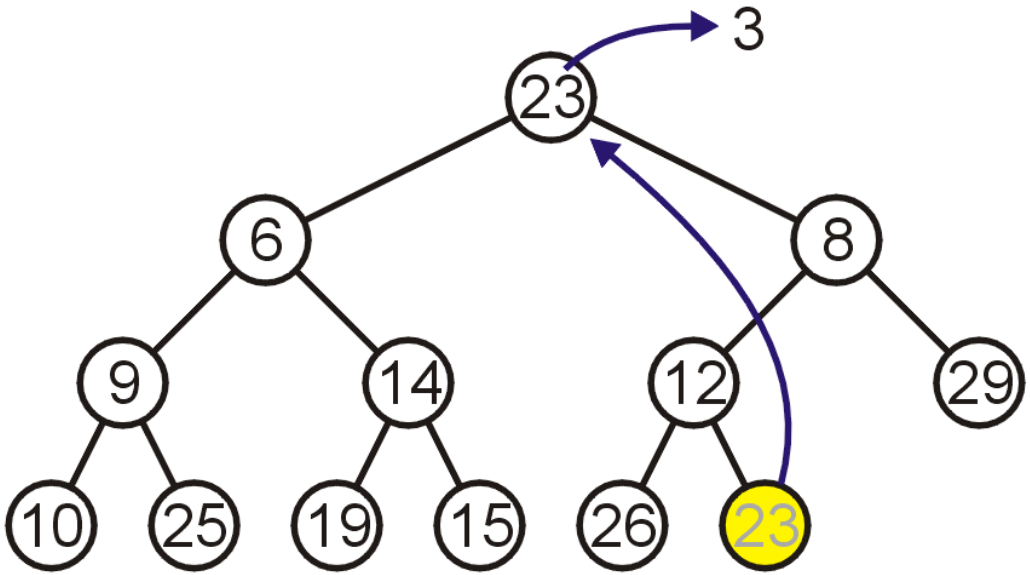
- At this point, 8 is greater than its parent, so we are finished

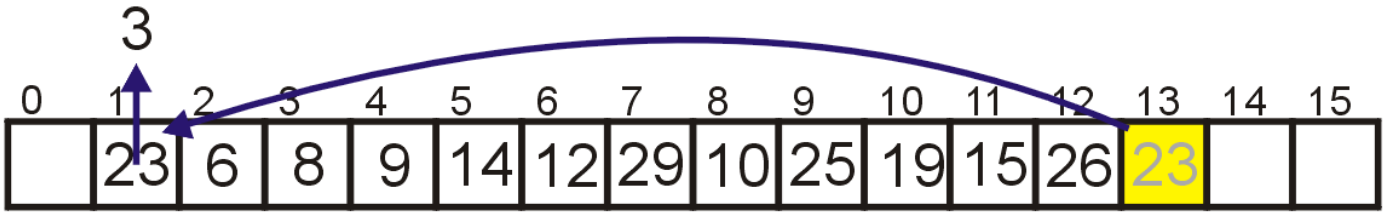



| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 3 | 6 | 8 | 9 | 14 | 12 | 29 | 10 | 25 | 19 | 15 | 26 | 23 | | |

# Array-Based Implementation – `deleteMin`

- Removing the top require copy of the last element to the top

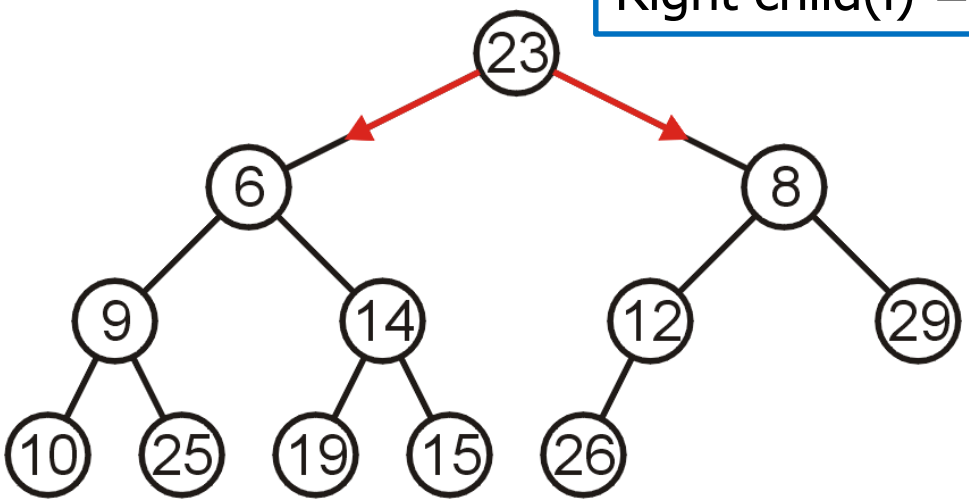


# Array-Based Implementation – `deleteMin`

- Percolate down
  - Compare Node 1 with its children: Nodes 2 and 3
  - Swap 23 and 6

Left child(i) = at position `2i`
Right child(i) = at position `2i + 1`

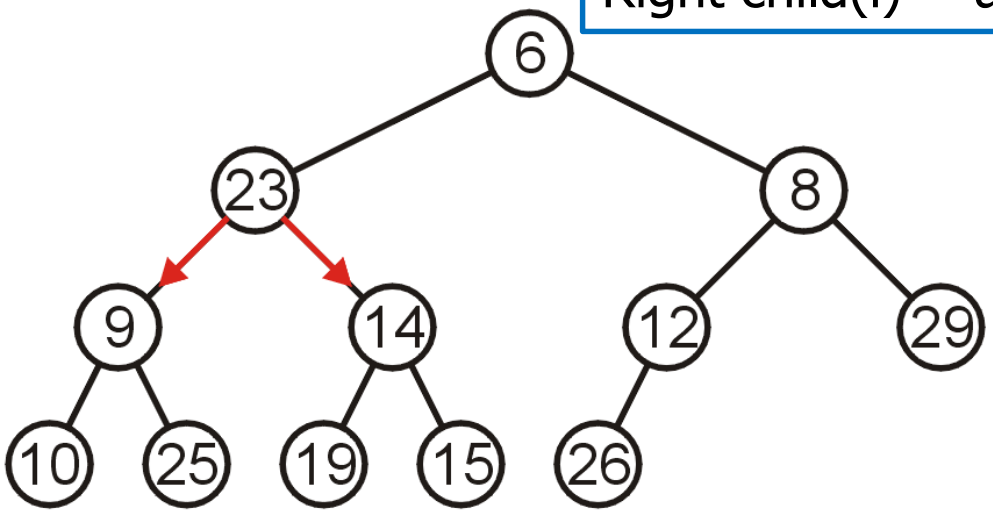



| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 23 | 6 | 8 | 9 | 14 | 12 | 29 | 10 | 25 | 19 | 15 | 26 | | | |

# Array-Based Implementation – `deleteMin`

- Compare Node 2 with its children: Nodes 4 and 5
  - Swap 23 and 9

Left child(i) = at position `2i`
Right child(i) = at position `2i + 1`

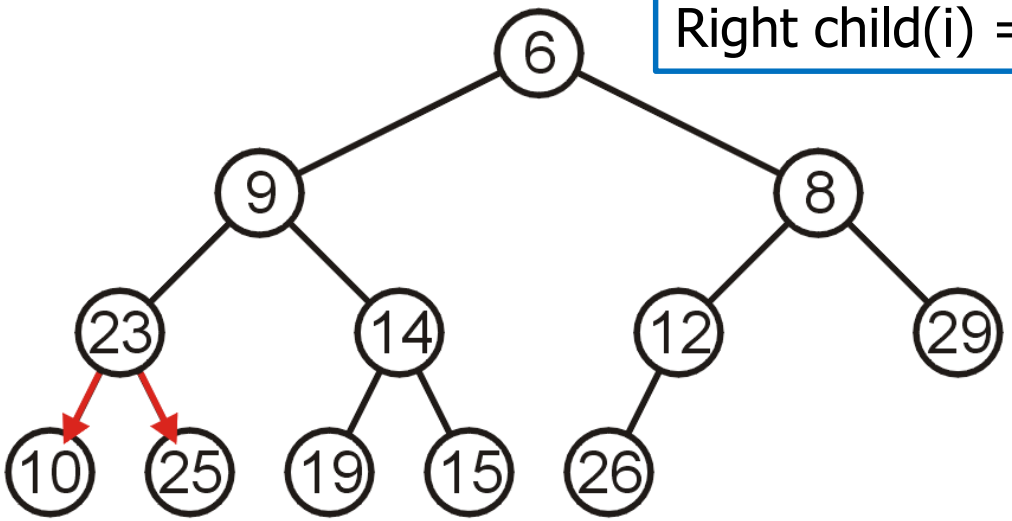



| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 6 | 23 | 8 | 9 | 14 | 12 | 29 | 10 | 25 | 19 | 15 | 26 | | | |

# Array-Based Implementation – `deleteMin`

- Compare Node 4 with its children: Nodes 8 and 9
  - Swap 23 and 10

Left child(i) = at position `2i`
Right child(i) = at position `2i + 1`

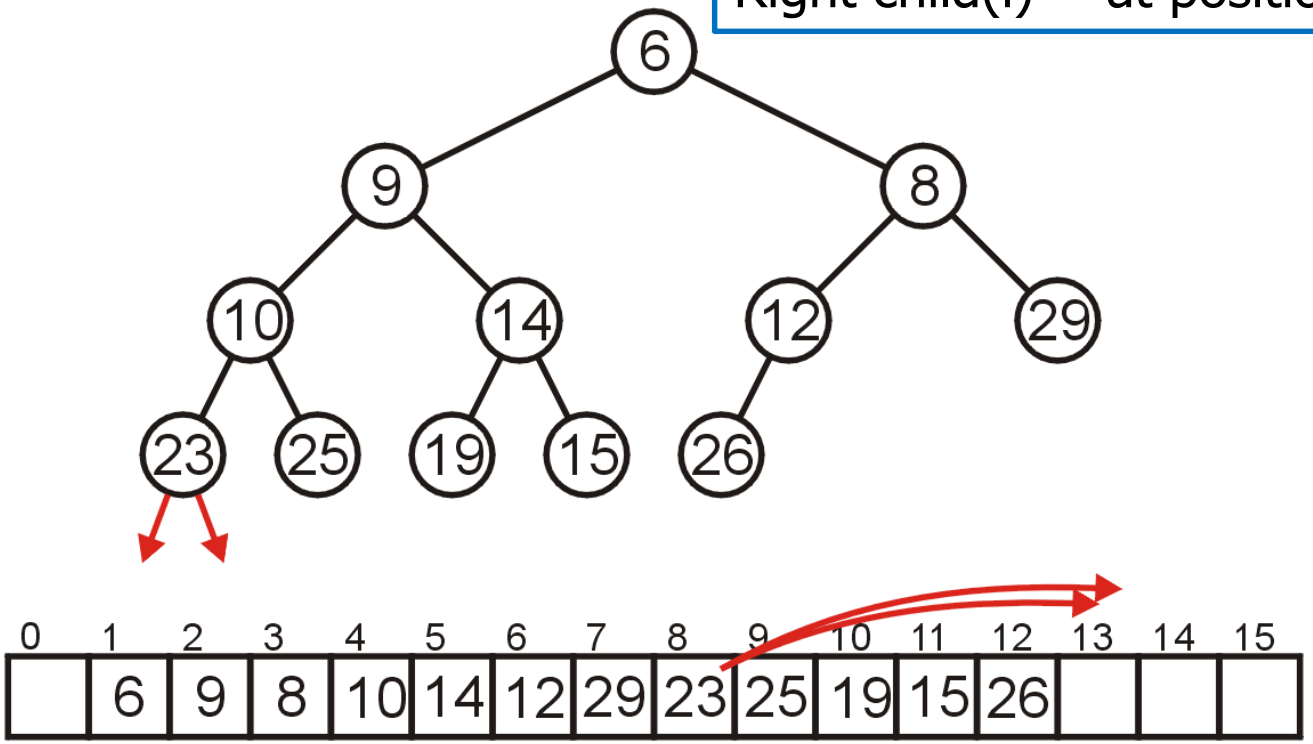



| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 6 | 9 | 8 | 23 | 14 | 12 | 29 | 10 | 25 | 19 | 15 | 26 | | | |

# Array-Based Implementation – `deleteMin`

- The children of Node 8 are beyond the end of the array:
  - Stop

Left child(i) = at position `2i`
Right child(i) = at position `2i + 1`



# Runtime Analysis

- `insert` operation
  - Worst case: Inserting an element less than the root
    - $O(\log_2 n)$
  - Best case: Inserting an element greater than any other element
    - $O(1)$
  - Average case: $O(1)$
    - Why ?

- `deleteMin` operation
  - Replacing the top element is $O(1)$
  - Percolate down the top object is $O(\log_2 n)$
  - We copy something that is already in the lowest depth
    - It will likely be moved back to the lowest depth

# Building a Heap

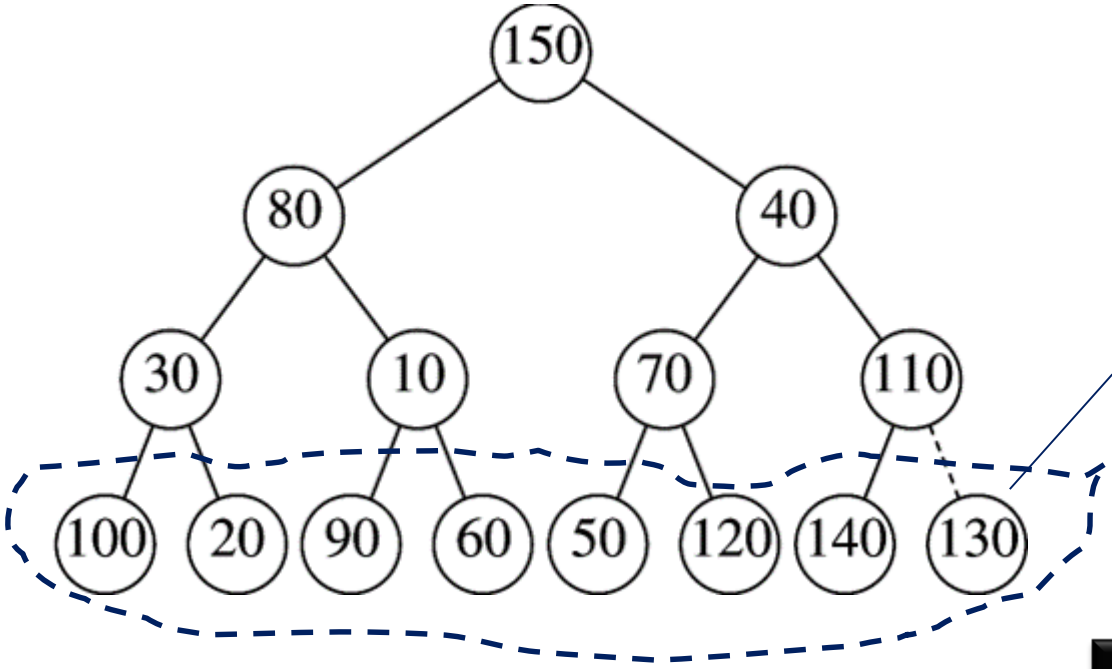
- What if all N elements are all available upfront?
  - Construct heap from initial set of N items

- Solution 1 (insert method)
  - Perform N inserts

- Solution 2 (BuildHeap method)
  - Randomly populate initial heap with structure property
  - Perform a percolate-down from each internal node
    - To take care of heap order property

# BuildHeap Example

- Input priority levels
  - { 150, 80, 40, 30, 10, 70, 110, 100, 20, 90, 60, 50, 120, 140, 130 }

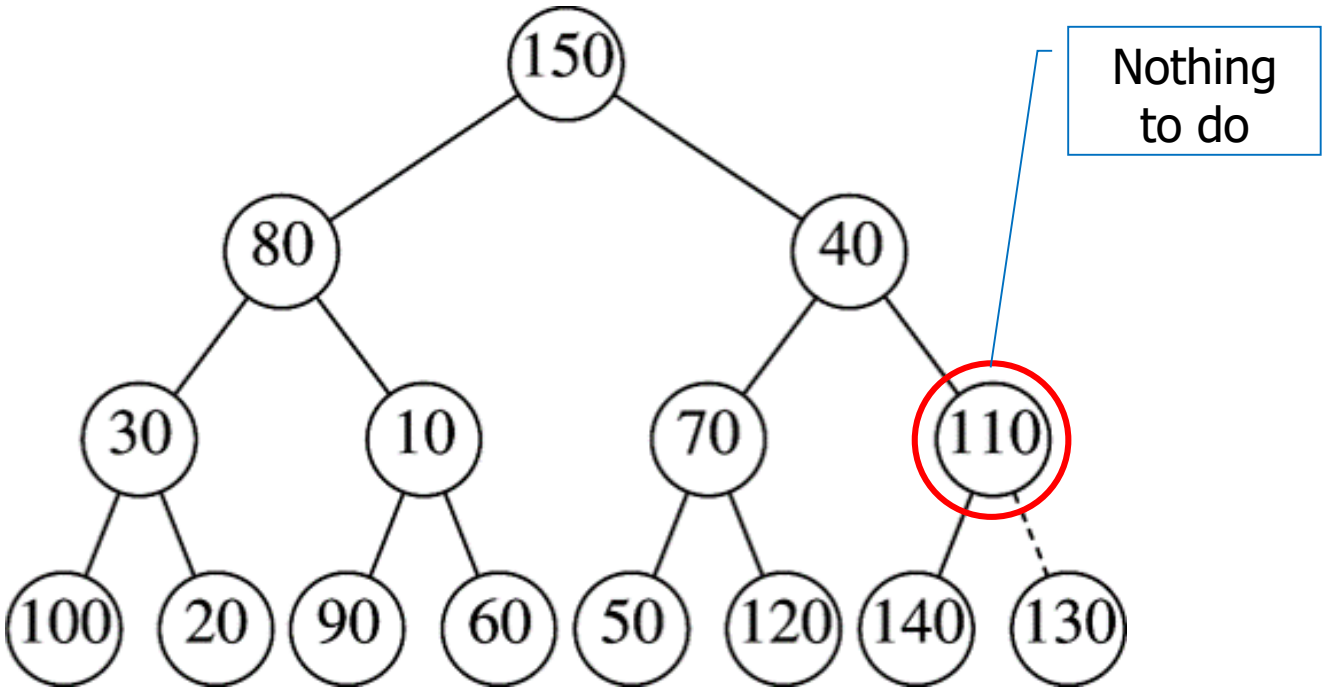



Leaves are all valid heaps (implicitly)

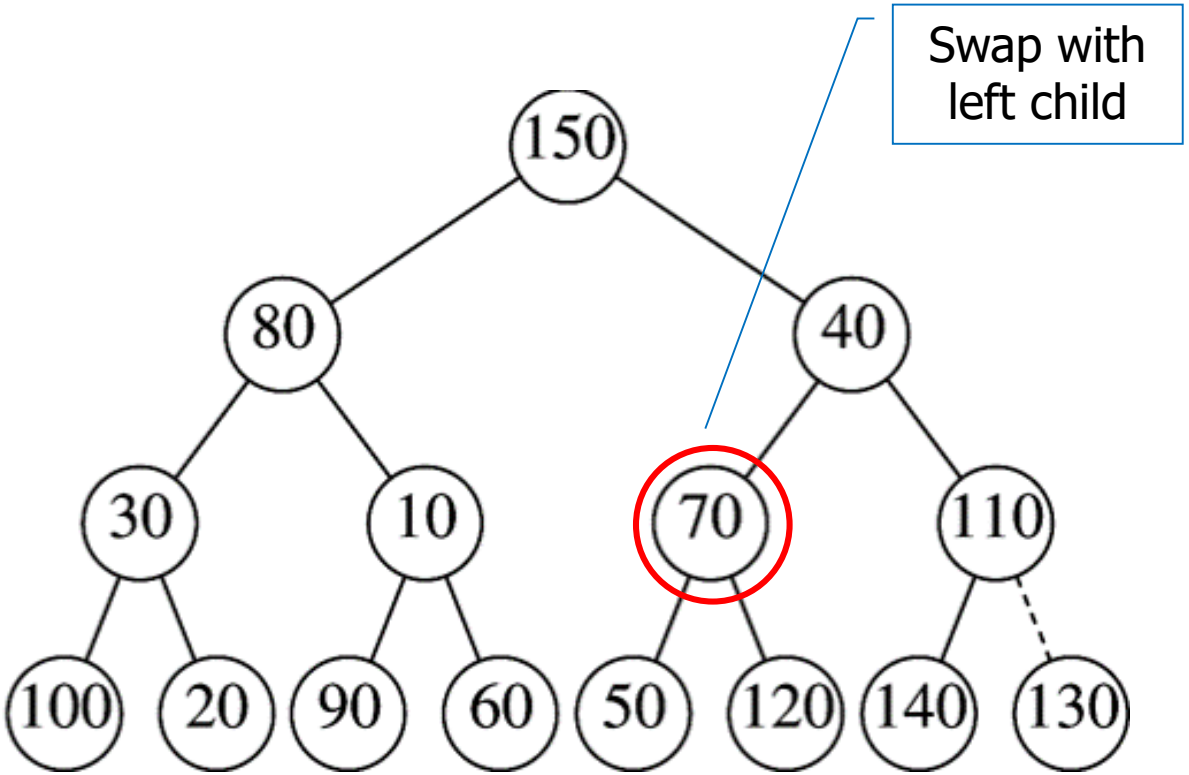
- Arbitrarily assign elements to heap nodes
- Structure property satisfied
- Heap order property violated
- Leaves are all valid heaps (implicit)

So, let us look at each internal node,
from bottom to top,
and fix if necessary

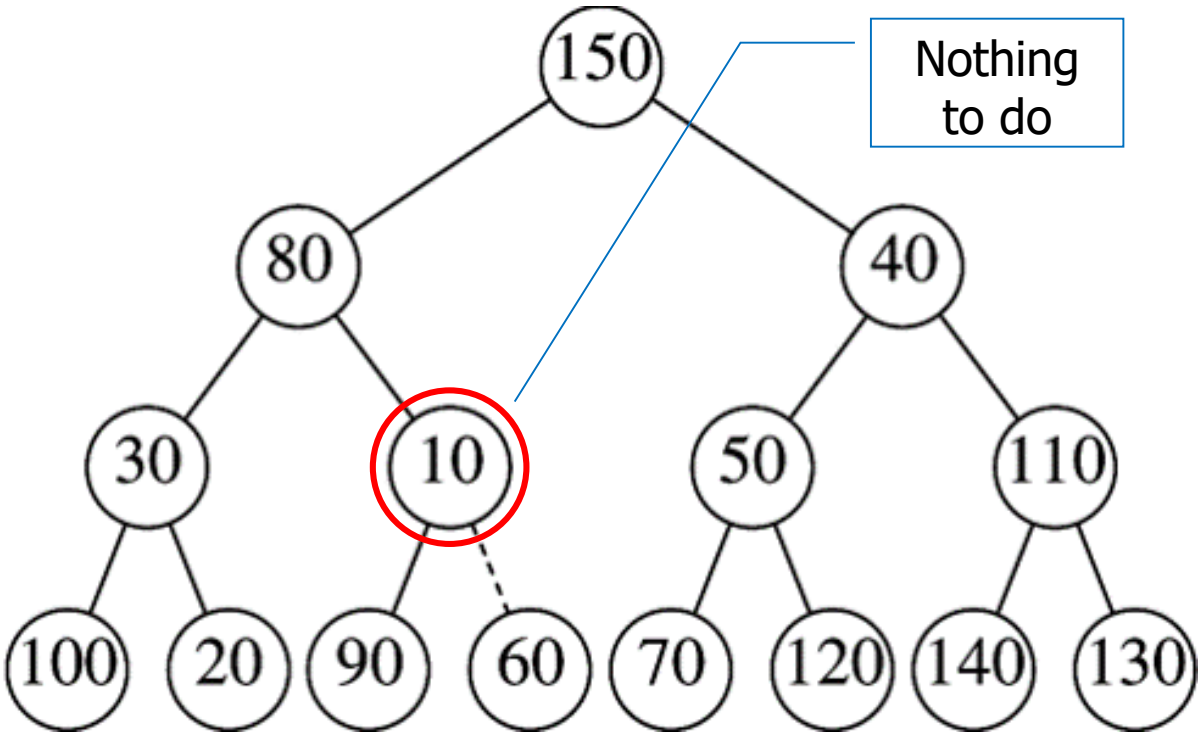
# BuildHeap Example



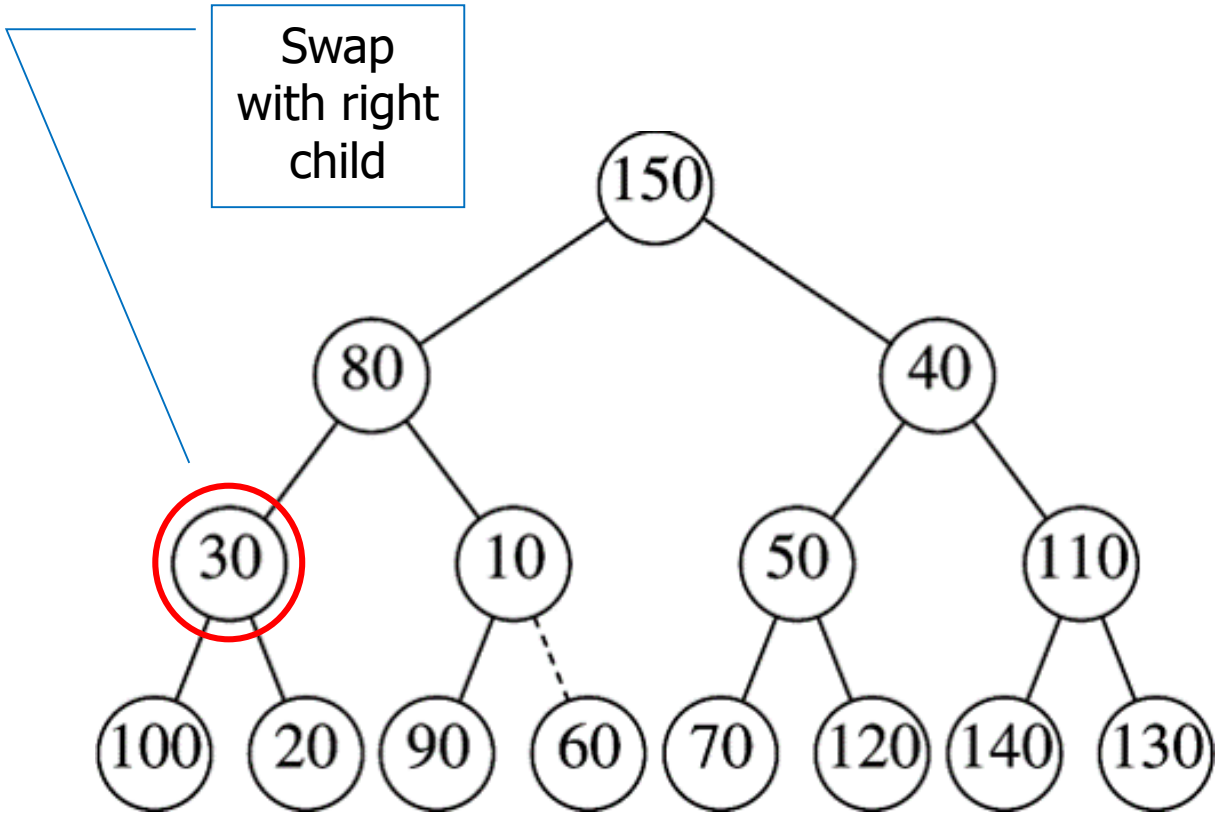
# BuildHeap Example



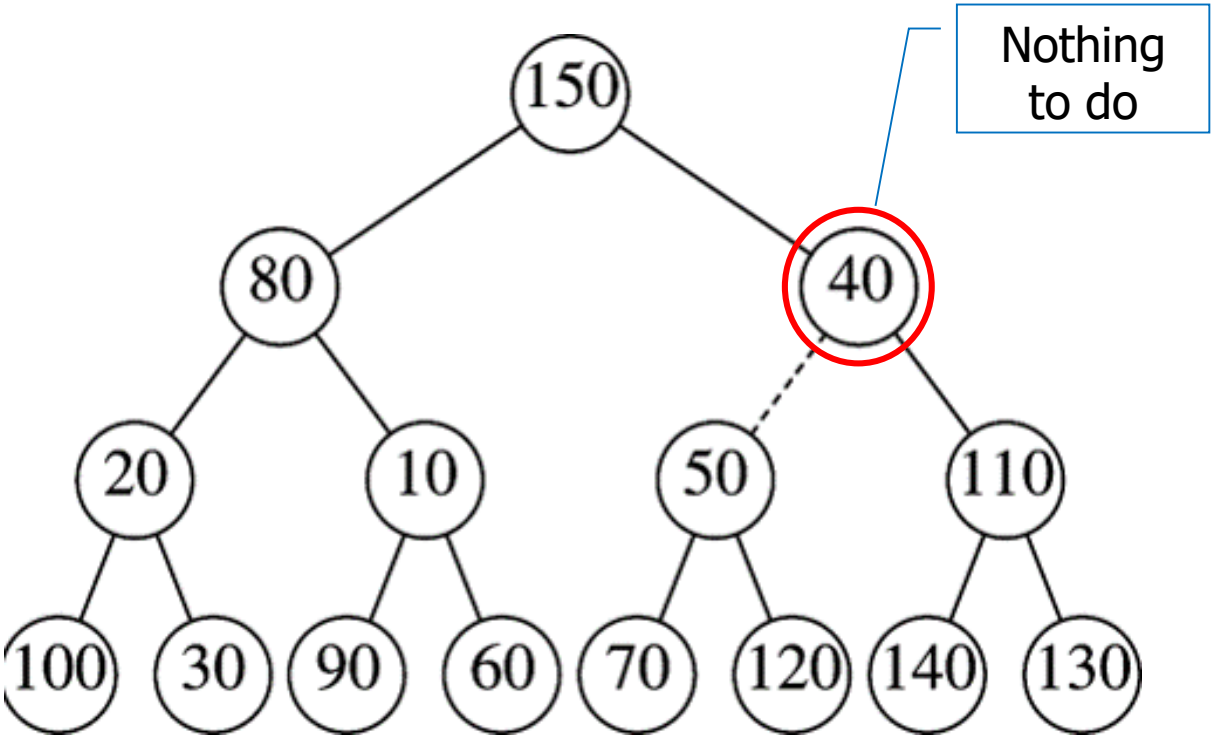
# BuildHeap Example



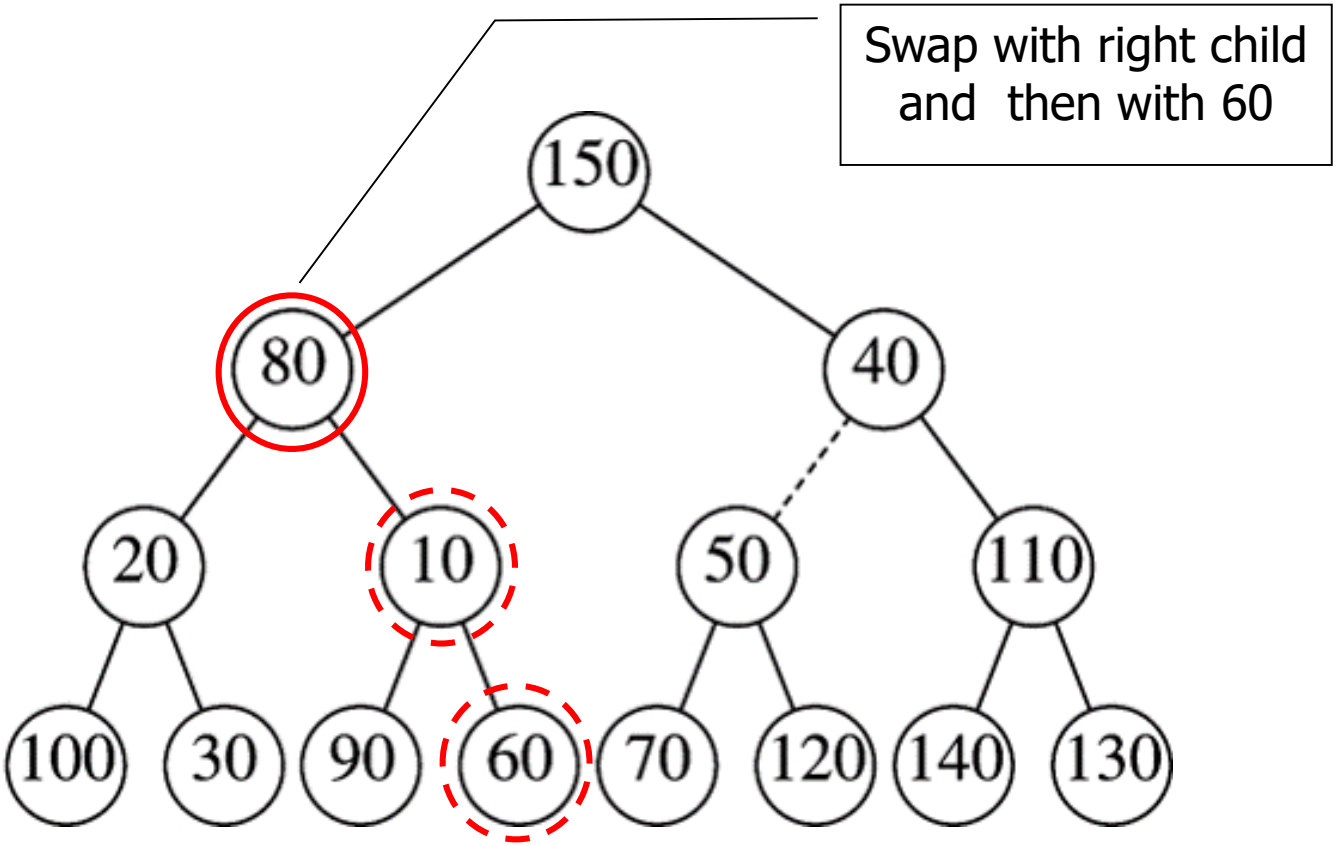
# BuildHeap Example



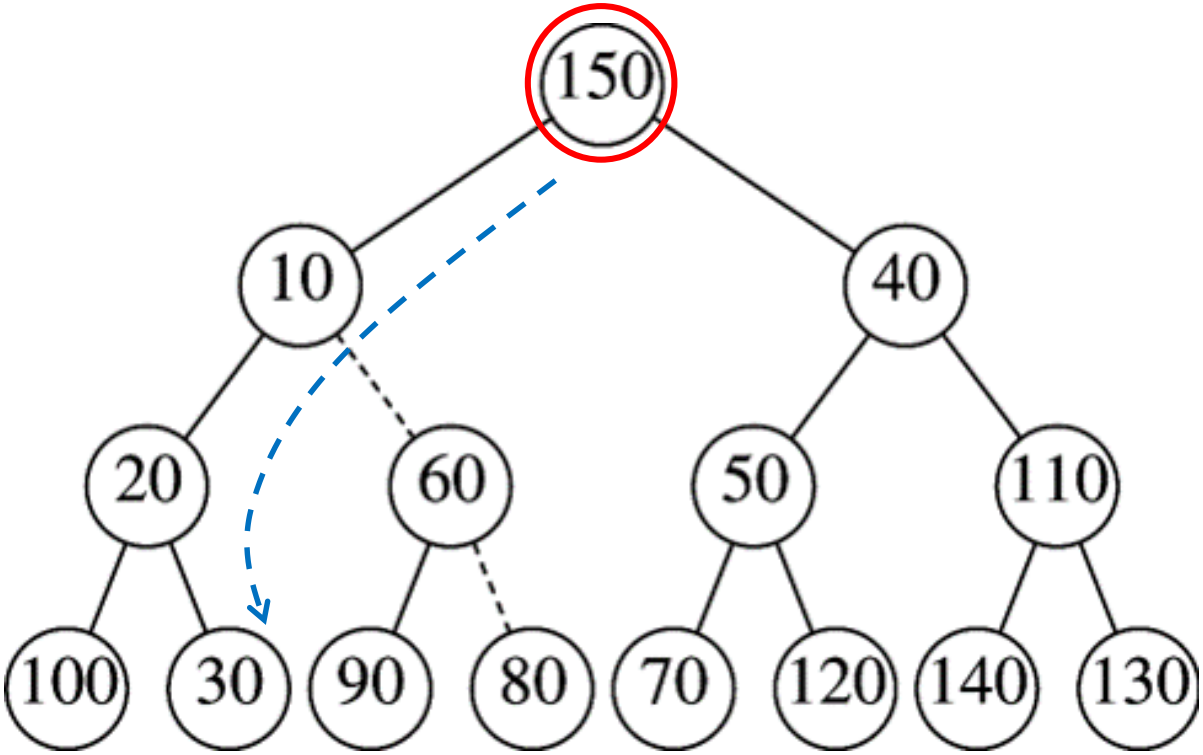
# BuildHeap Example



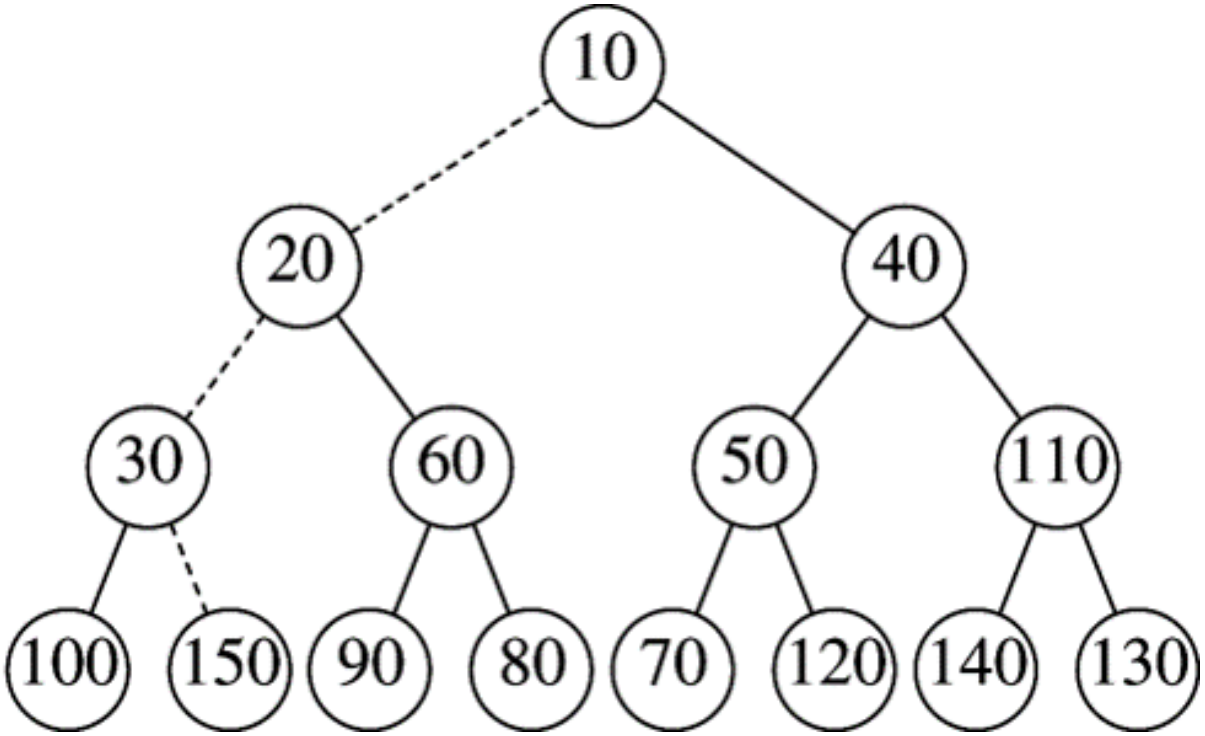
# BuildHeap Example



# BuildHeap Example



# BuildHeap Example




Final Heap

# Any Question So Far?

# Practice Exercise 1

- Write a recursive function isHeap that reads in the level-order (see slide notes) traversal of a binary tree from a file and returns true if the tree is a valid binary min-heap, i.e. the min-heap property is satisfied.
- Heap property: the root is smaller than its left and right child, and both left and right subtrees are min heaps too.
- Test your code by passing in a binary tree that is a min-heap and another binary tree that is not a min-heap.
- Submit (a) your code and both files and (b) a report that includes:
- A screenshot of the first (min-heap) file you pass
- A screenshot of the output after running the code for the first file
- A screenshot of the second (not min-heap) file you pass
- A screenshot of the output after running the code for the second file
- You can use an array-based or pointer-based implementation of a tree.

# Practice Exercise 2

- Write code to insert an integer sequence read from a file into an initially empty min-heap through (a) the build heap method and (b) the insert method.
- Show the result of 2 delete-min operations on the min-heap, one by one.
- You can use an array-based implementation of a min-heap.
- Submit your file along with your code. Your code should run with any other txt file that contains an integer sequence.
- Also include this exercise in your report, i.e. screenshots of the file you pass, the output after build heap and insert methods and the screenshot of the heap after each delete-min operation.